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Power BI Tutorial

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1. Project 2 Overview: Inventory Data Analysis

This project focuses on analyzing inventory data to derive meaningful insights related to demand, availability, shortages, profit, and loss using Power BI. The objective is to build an interactive and dynamic report that helps stakeholders make informed decisions regarding inventory management.

The report is designed across two pages, each highlighting key performance indicators (KPIs):

- **Page 1 (Inventory Insights):**

This page presents critical inventory metrics such as Average Demand Per Day, Average Availability Per Day, and Total Supply Shortage using card visuals. A Filter Pane is incorporated to allow users to dynamically explore and analyze the data based on their requirements.

- **Page 2 (Financial Insights):**

This page focuses on financial performance, displaying Total Profit, Total Loss, and Average Daily Loss through intuitive card visuals. Similar to the first page, a Filter Pane is provided to enhance user interaction and flexibility.

The data pipeline begins with connecting **Power BI to Microsoft SQL Server**, where initial analysis is performed on a test dataset. DAX measures and KPIs are created and validated in the test environment before transitioning to the production dataset.

To simulate real-world business scenarios, the project also demonstrates data source migration. The data source is switched from SQL Server to MySQL while maintaining the same DAX calculations, ensuring consistency and reliability of the report across different database systems.

This project highlights key skills in:

- ✓ Data integration and transformation
- ✓ DAX-based KPI development
- ✓ Interactive dashboard design
- ✓ Handling test and production environments
- ✓ Data source migration and validation

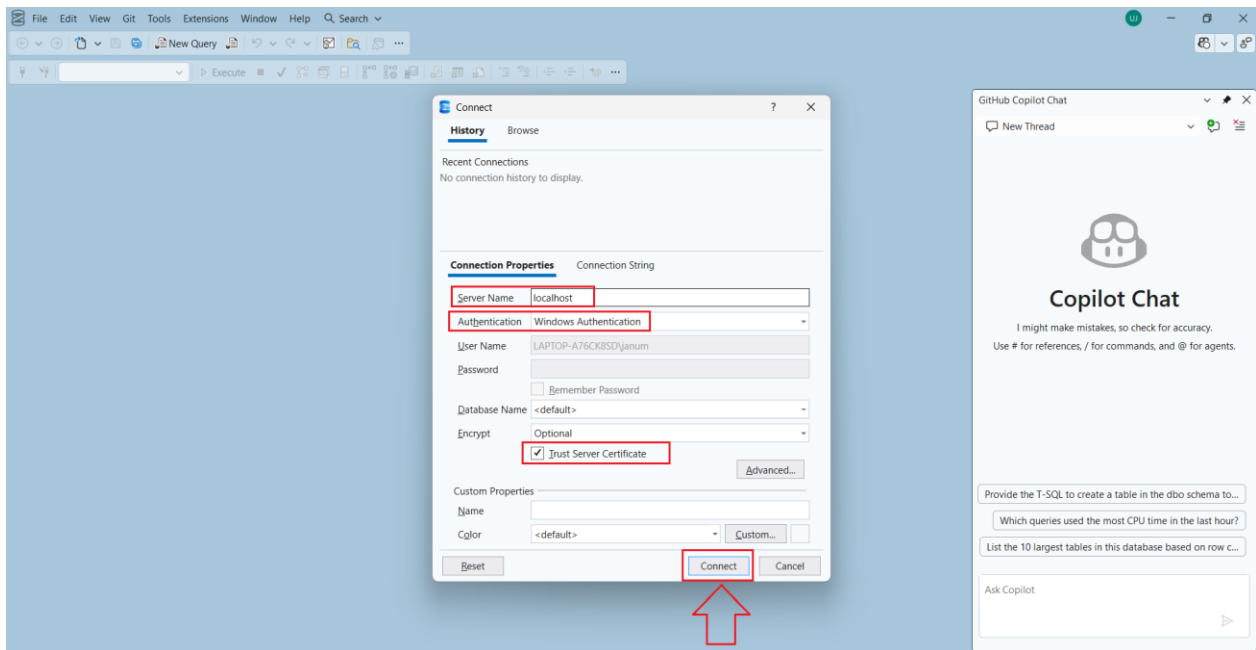
Overall, the project showcases a practical approach to building scalable and flexible inventory analytics solutions using Power BI.

2. Connecting Power BI with Microsoft SQL Server

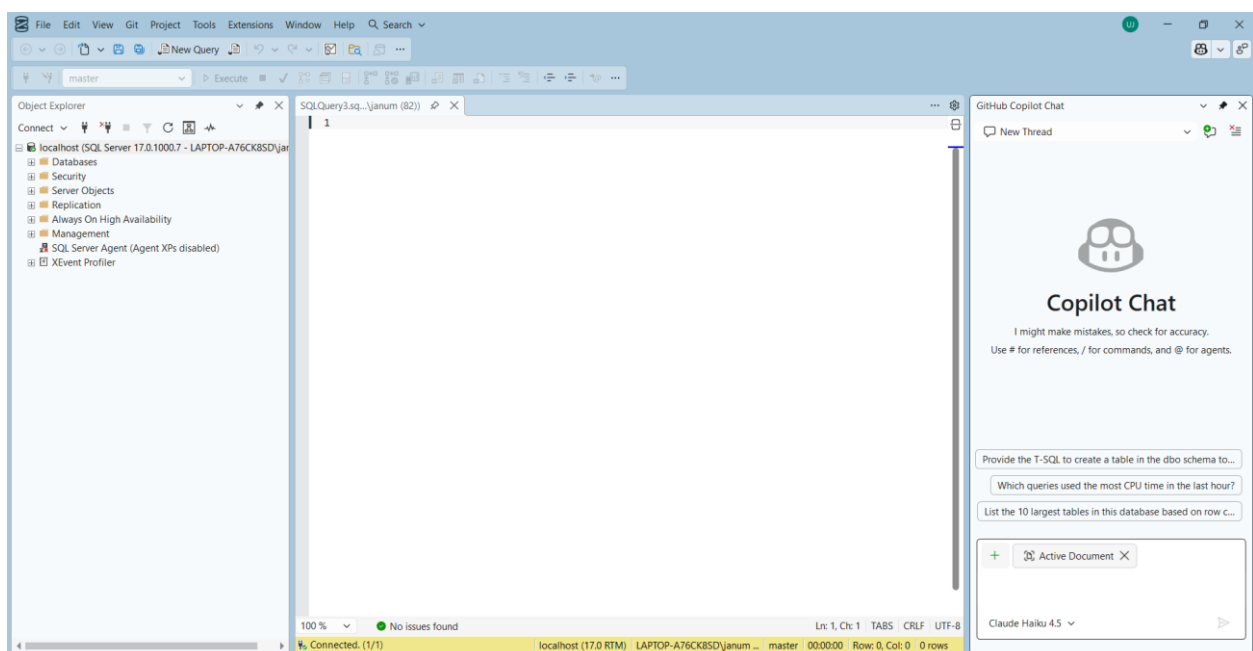
2.1. Download Microsoft SQL Server

SQL Server Standard Developer Edition: <https://www.microsoft.com/en-in/sql-server/sql-server-downloads>

Install SSMS: <https://learn.microsoft.com/en-us/ssms/install/install?redirectedfrom=MSDN>



Now I can Simply Use it



2.2. Plan of Action

Our plan is to start with a 'Test env' and then eventually move to the 'Production env'.

- **Test Environment:** A testing setup used for development, experimentation, and validation of reports. It allows us to check data, logic, and transformations without affecting real-world operations.
- **Production Environment:** The live environment where finalized reports are deployed and used by end users for decision-making. The data here is real and should not be altered unnecessarily.

- Changing 'Test env' → 'Production env in Same Database

In Power BI

Transitioning the report source from the **Test environment** to the **Production environment** within the same database can be done in two ways:

i. Using Data Source Settings

- Go to **Transform Data → Data Source Settings**
- Select the current data source
- Click **Change Source**
- Update the server/database details to the Production Environment

ii. Using Power Query Editor

- Open **Transform Data → Power Query Editor**
- Select the required query
- Locate and modify the **Source** step (change server name, database name, or environment)
- Click **Close & Apply** to apply the changes

2.3. Practical – SQL Server

- Test Environment

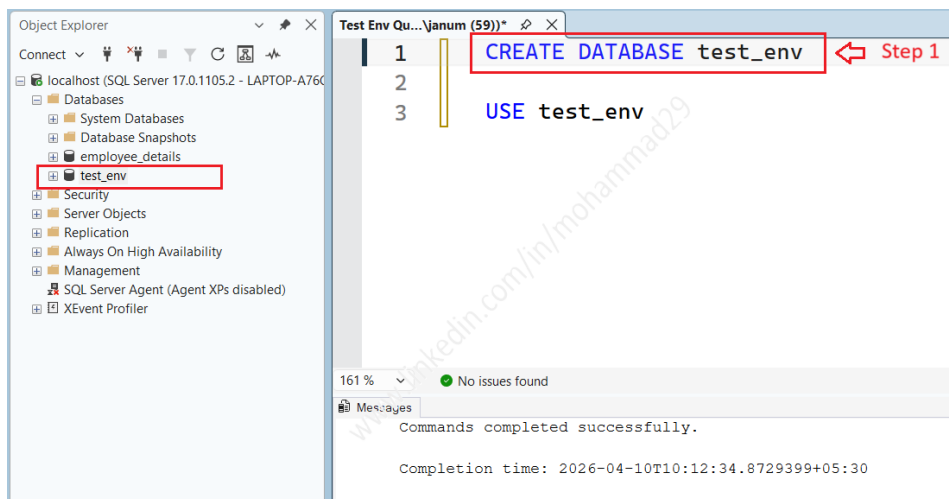
i. Test Dataset

We have few tables:

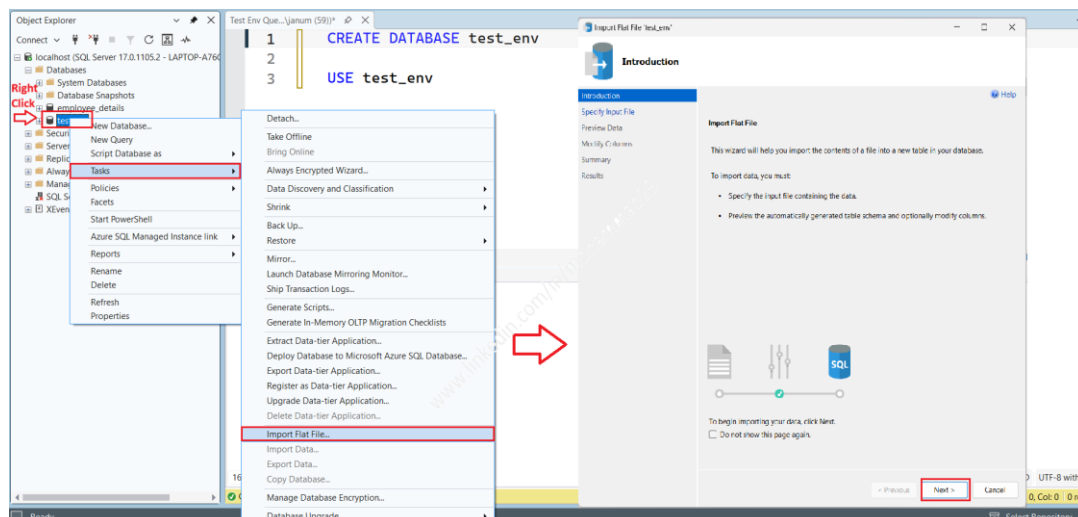
1. Demand & Availability Table
2. Products Table
3. KPI Sheet (that we have to represent on the PowerBI report)

ii. Importing the Test-Data in Test-Environment in SQL Server

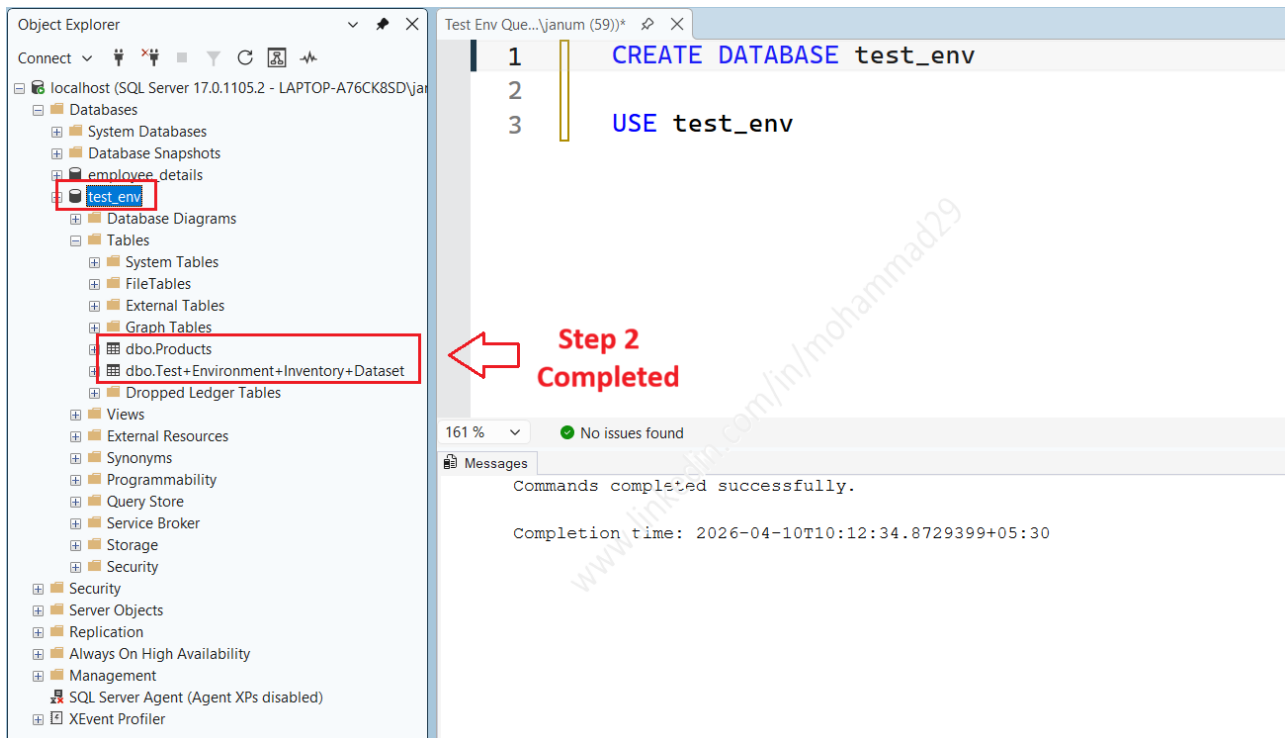
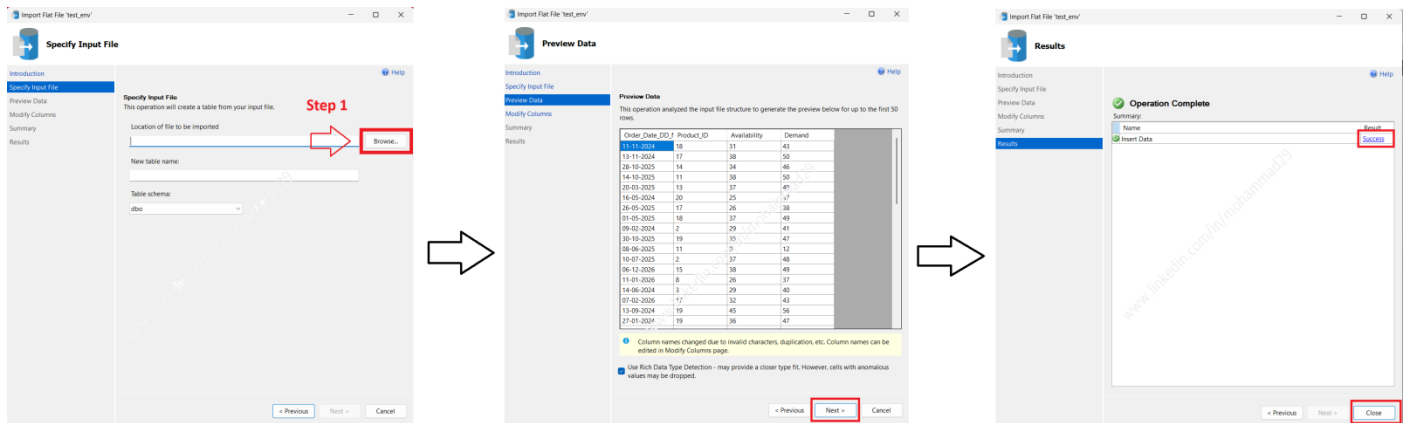
Step 1: Create a new database by the name test_env database in SQL Server.



Step 2: Once the Database is created, import the test-data in the test_env.



Day 7 - Power BI for Data Analysis



Day 7 - Power BI for Data Analysis

Step 3: Applying Left Join in SQL Server to prepare the data to be used for reporting.

Products Table

	Product_ID	Product_Name	Unit_Price
1	1	Wireless Mouse	15.9899997711182
2	2	Bluetooth Headphones	39.9900016784668
3	3	4K Smart TV	499.989990234375
4	4	Portable Charger	25.4899997711182
5	5	Gaming Keyboard	79.9899978637695
6	6	Laptop Stand	29.9899997711182
7	7	Phone Case	12.9899997711182
8	8	Digital Camera	299.989990234375
9	9	Smartwatch	199.990005493164
10	10	Tablet	249.990005493164
11	11	Noise-Canceling Hea...	89.9899978637695
12	12	Electric Toothbrush	49.9900016784668
13	13	Air Purifier	119.98999786377
14	14	Coffee Maker	59.9900016784668

Test+Environment+Inventory+Dataset

	Order_Date_DD_MM_YYYY	Product_ID	Availability	Demand
1	2024-11-11	18	31	43
2	2024-11-13	17	38	50
3	2025-10-28	14	34	46
4	2025-10-14	11	38	50
5	2025-03-20	13	37	49
6	2024-05-16	20	25	37
7	2025-05-26	17	26	38
8	2025-05-01	18	37	49
9	2024-02-09	2	29	41
10	2025-10-30	19	35	47
11	2025-06-08	11	0	12
12	2025-07-10	2	37	48
13	2026-12-06	15	38	49
14	2026-01-11	8	26	37

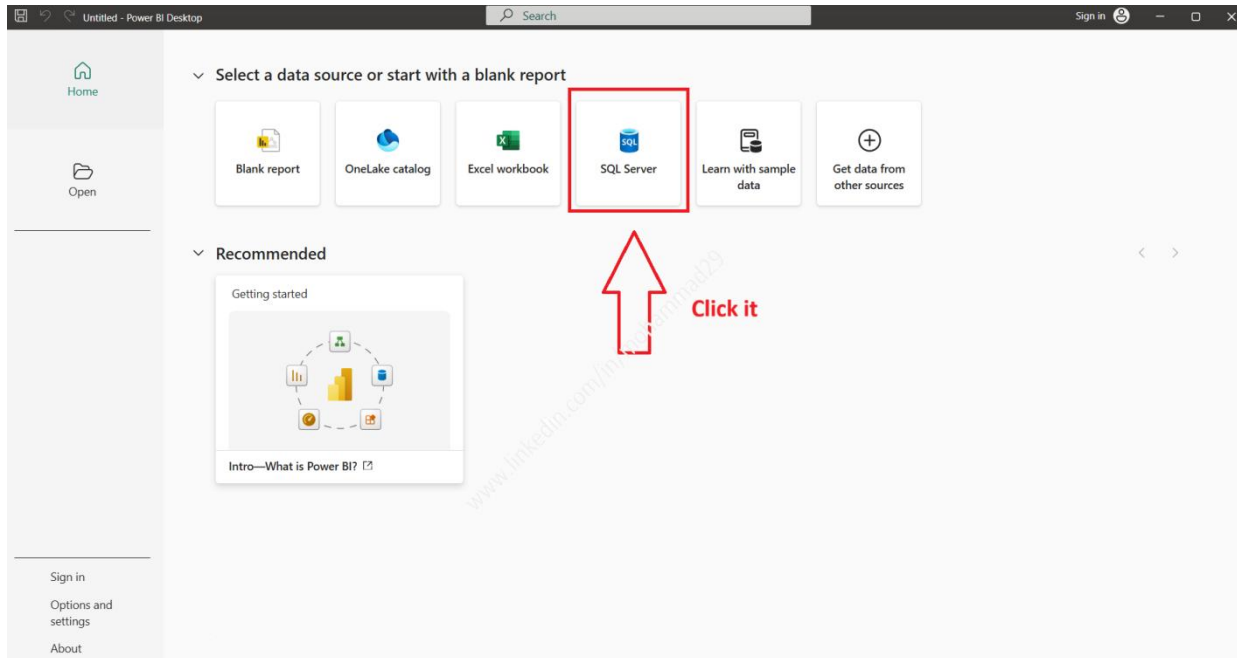
Left Join

new_table

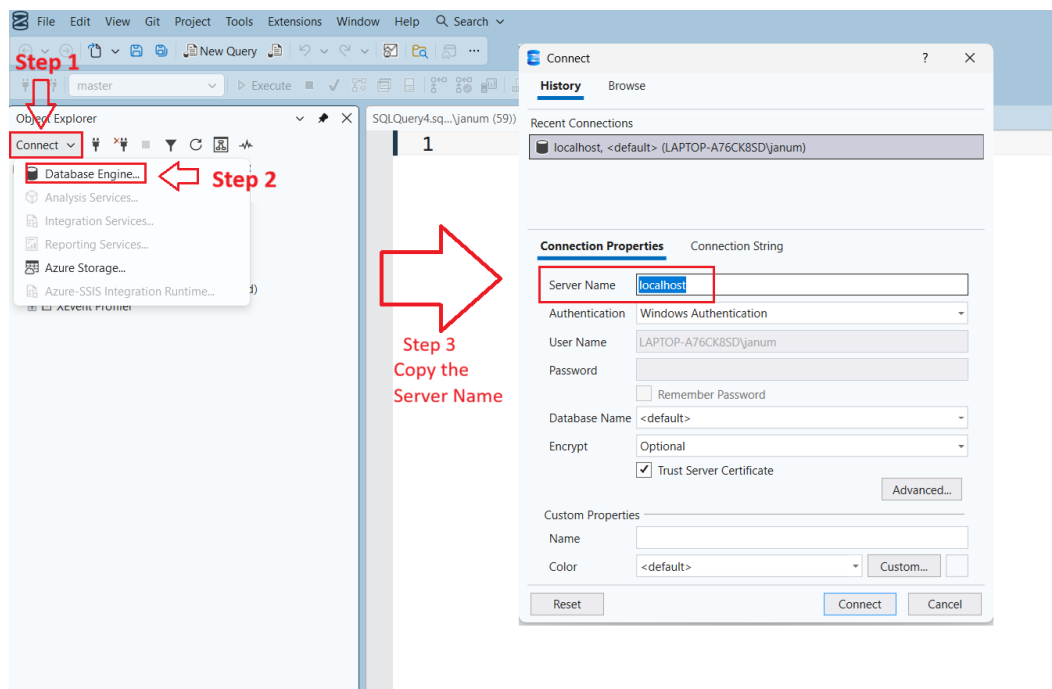
	Order_Date_DD_MM_YYYY	Product_id	availability	demand	product_name	unit_price
1	2024-11-11	18	31	43	VR Headset	399.989990234375
2	2024-11-13	17	38	50	Action Camera	169.990005493164
3	2025-10-28	14	34	46	Coffee Maker	59.9900016784668
4	2025-10-14	11	38	50	Noise-Canceling Headphones	89.9899978637695
5	2025-03-20	13	37	49	Air Purifier	119.98999786377
6	2024-05-16	20	25	37	Smart Thermostat	129.990005493164
7	2025-05-26	17	26	38	Action Camera	169.990005493164
8	2025-05-01	18	37	49	VR Headset	399.989990234375
9	2024-02-09	2	29	41	Bluetooth Headphones	39.9900016784668
10	2025-10-30	19	35	47	Bluetooth Speaker	39.9900016784668
11	2025-06-08	11	0	12	Noise-Canceling Headphones	89.9899978637695
12	2025-07-10	2	37	48	Bluetooth Headphones	39.9900016784668
13	2026-12-06	15	38	49	LED Desk Lamp	18.9899997711182
14	2026-01-11	8	26	37	Digital Camera	299.989990234375

iii. Importing the New-Table to Power BI Desktop from in Test Environment

Step 1: Open Power BI Desktop and select SQL Server as the data source.

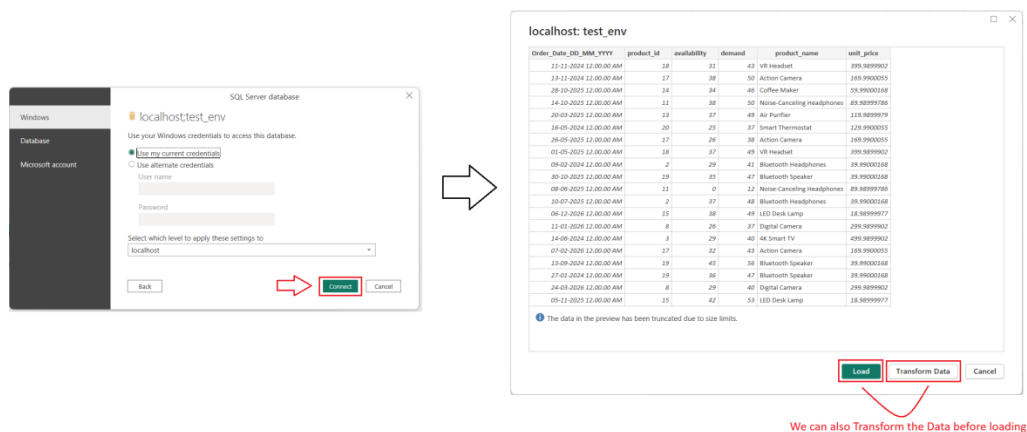
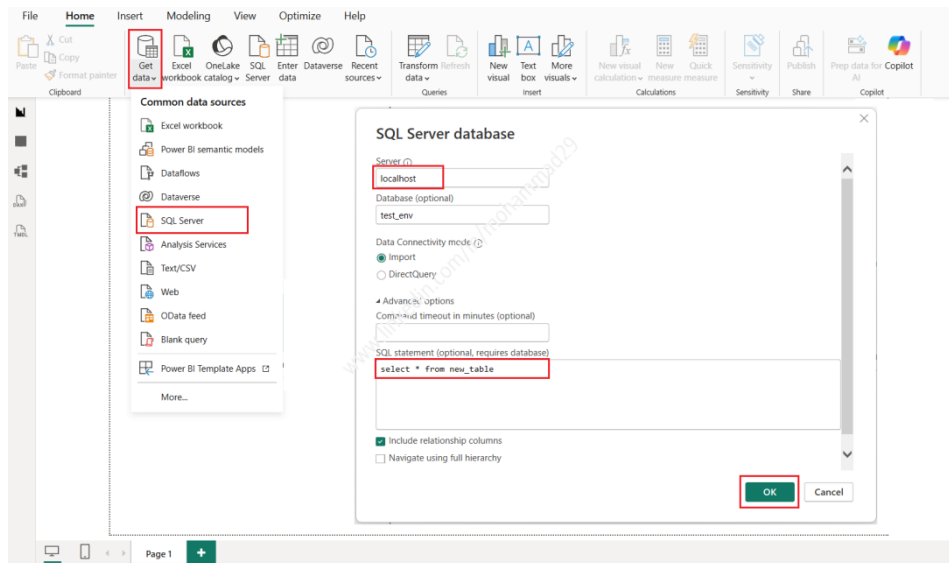


Step 2: Copy the Server name from SQL Server

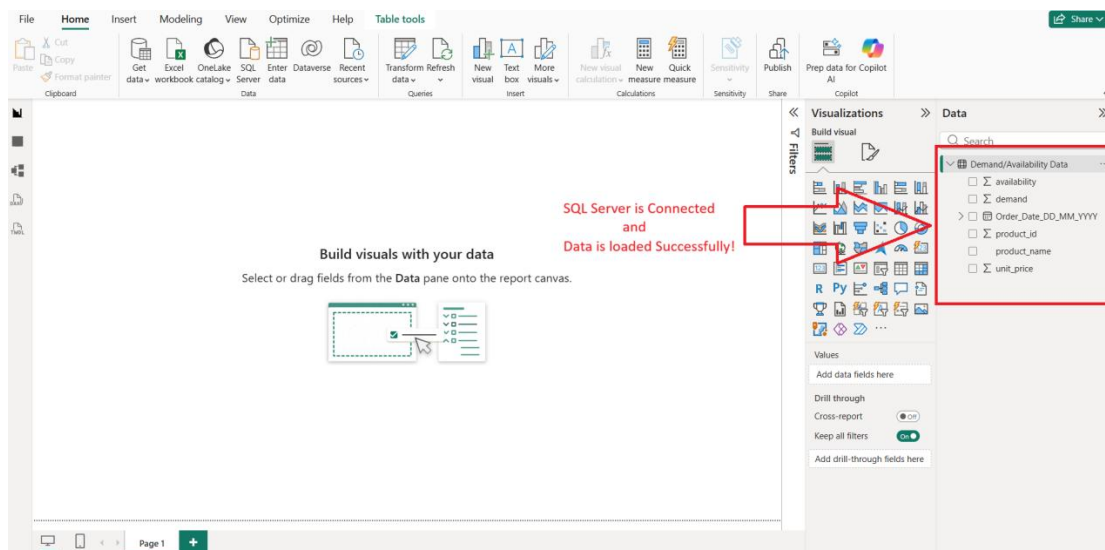


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Step 3: Connect SQL Server to Power BI using the server name, then load the test data.

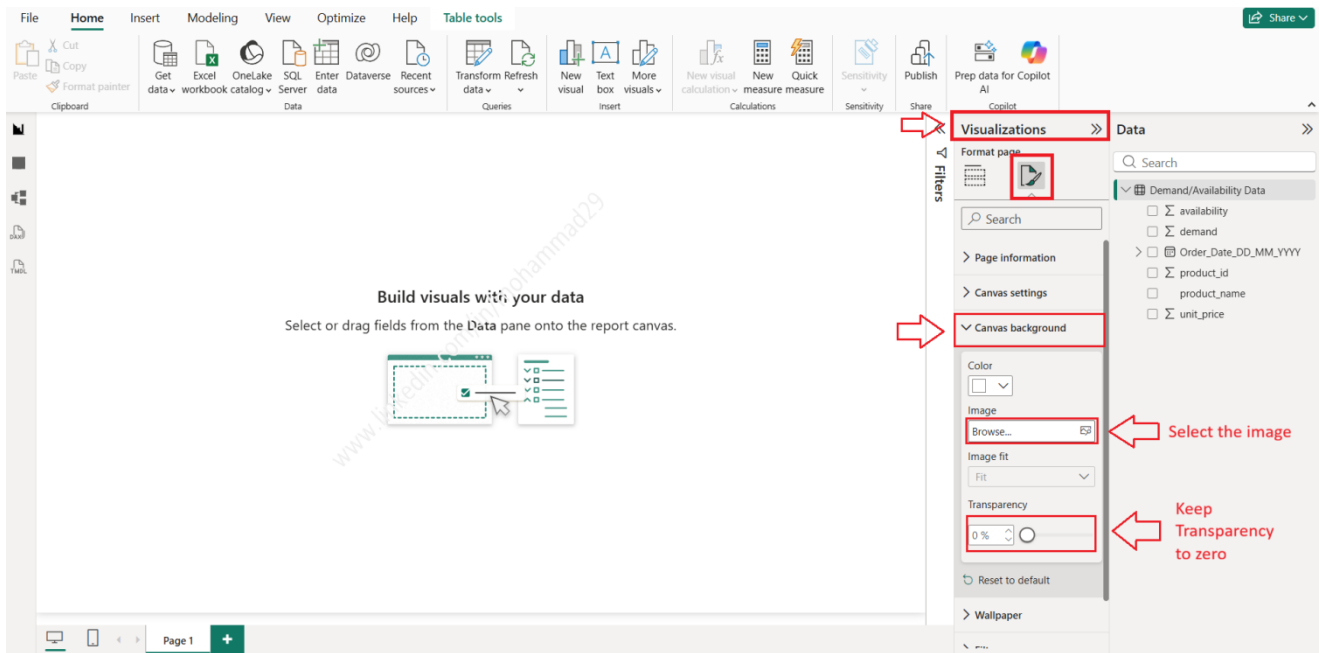


We can also Transform the Data before loading

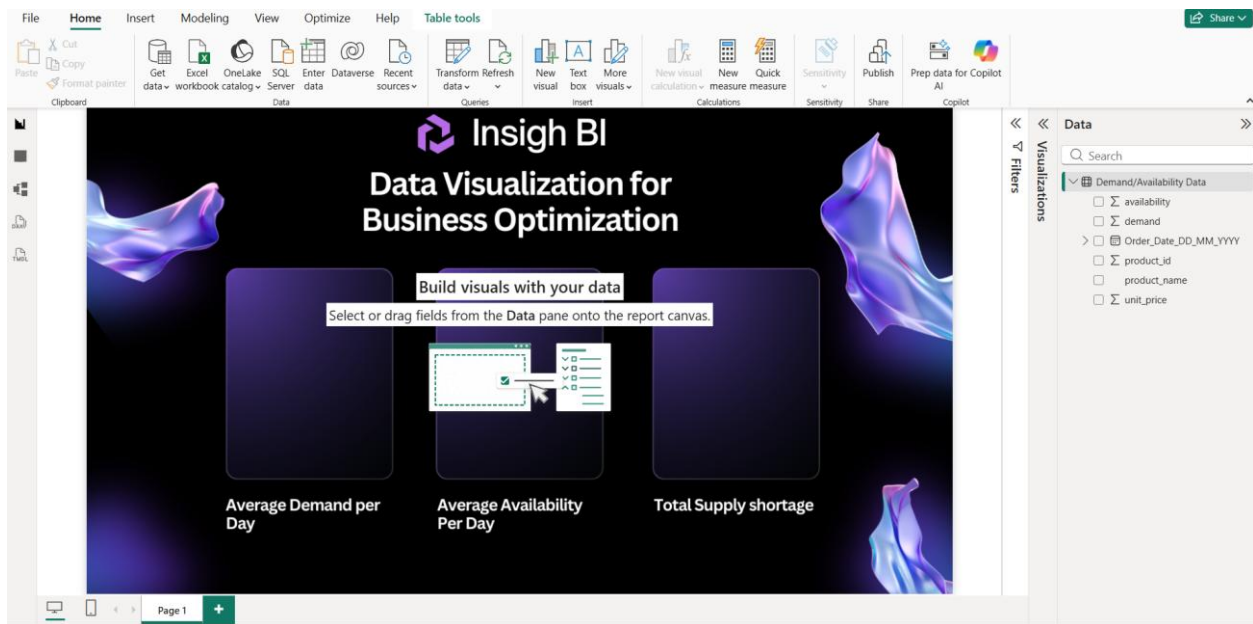


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Step 4: We can also add a background (designed in Canva) to the report page we are going to create.

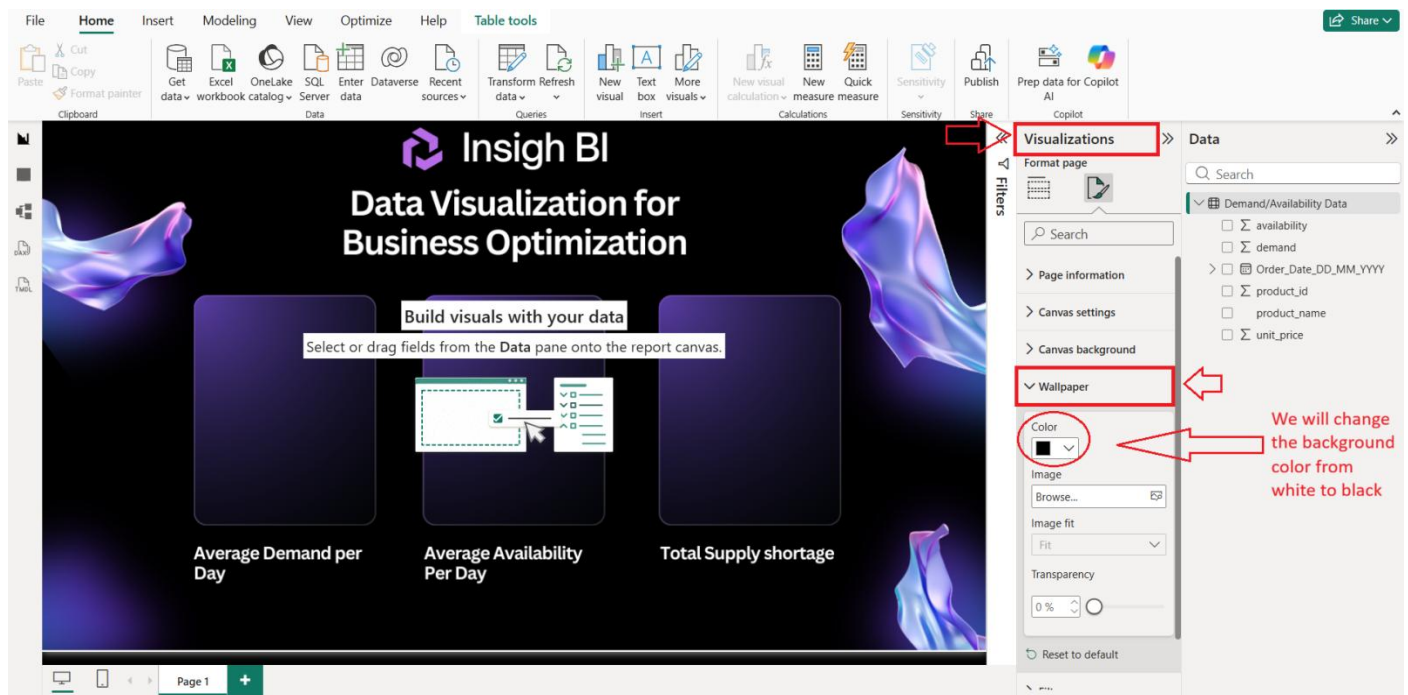


It's first look will be like this,



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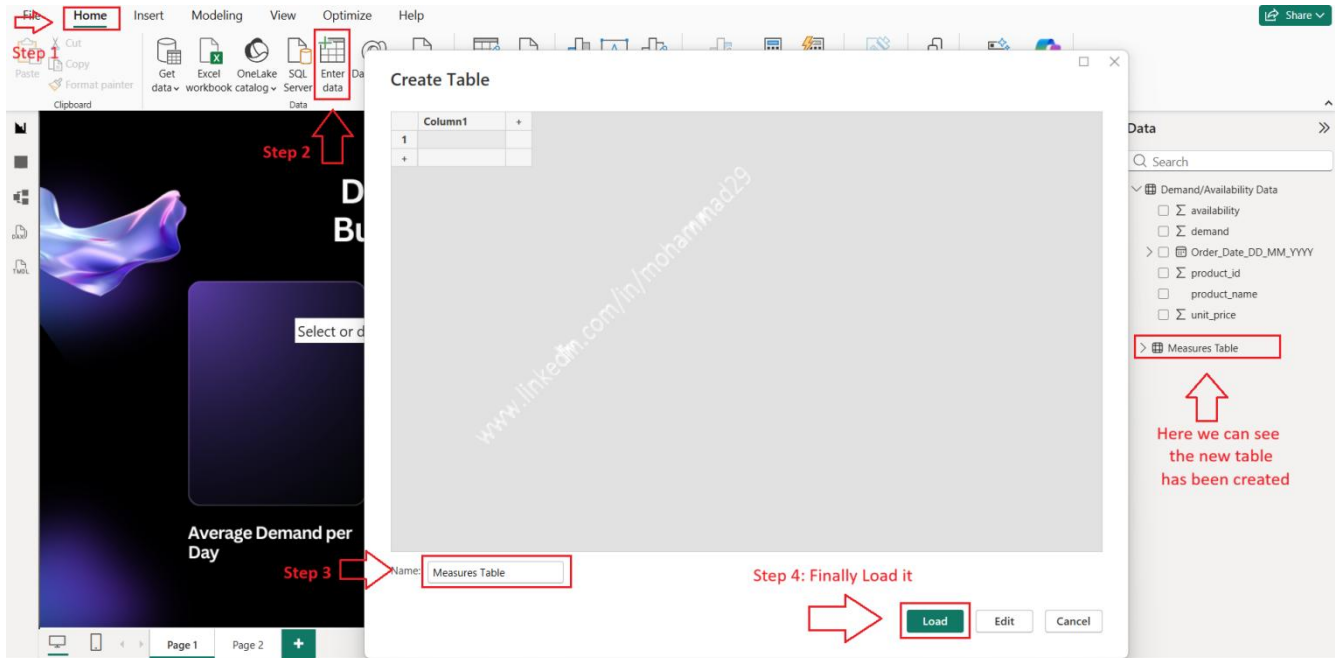
Let's say we want to change the color of the wallpaper as well.



Similarly we can do it for page 2.

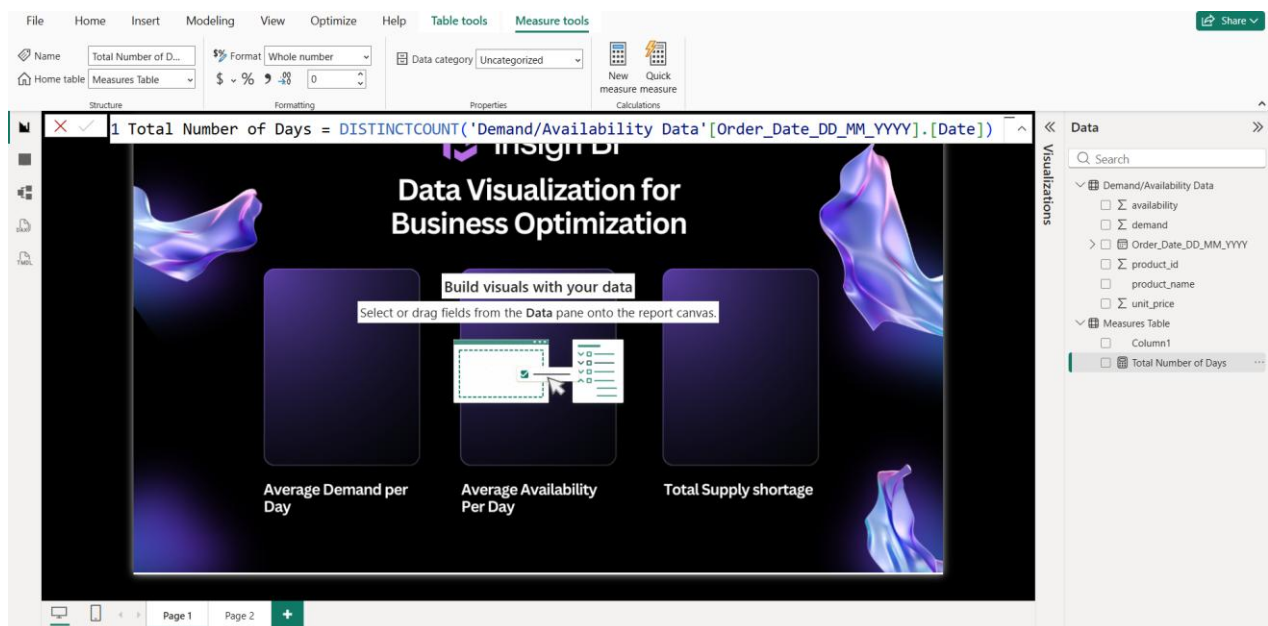
iv. Creating the DAX Measures & KPI

Step 1: First of all, let's create a separate table for Measures.



Step 2: In the measures table, let's create few measure:

i. Measure find out the distinct count of date values in the date column.



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ii. Measure to find out the Total Demand

The screenshot shows the Power BI Desktop interface. The 'Measure tools' tab is active. The 'Name' field is set to 'Total Demand' and the 'Format' is set to 'Whole number'. The 'Data category' is set to 'Uncategorized'. The 'Home table' is set to 'Measures Table'. The 'Measure' field contains the formula: `1 Total Demand = SUM('Demand/Availability Data'[demand])`. The 'Data' pane on the right shows the 'Demand/Availability Data' table with fields: `Σ availability`, `Σ demand`, `Order_Date_DD_MM_YYYY`, `Σ product_id`, `product_name`, `Σ unit_price`. The 'Measures Table' is also shown with fields: `Column1`, `Total Demand`, and `Total Number of Days`. The report canvas displays a 'Data Visualization for Business Optimization' dashboard with three visual placeholders: 'Average Demand per Day', 'Average Availability Per Day', and 'Total Supply shortage'.

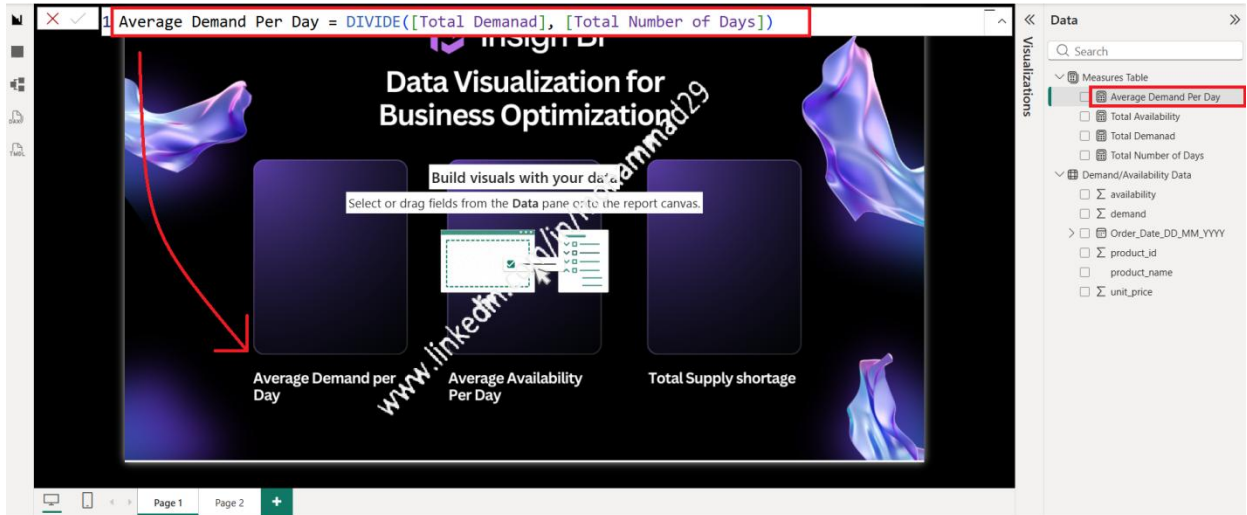
iii. Measure to find out the Total Availability

The screenshot shows the Power BI Desktop interface. The 'Measure tools' tab is active. The 'Name' field is set to 'Total Availability' and the 'Format' is set to 'Whole number'. The 'Data category' is set to 'Uncategorized'. The 'Home table' is set to 'Measures Table'. The 'Measure' field contains the formula: `1 Total Availability = SUM('Demand/Availability Data'[availability])`. The 'Data' pane on the right shows the 'Measures Table' with fields: `Σ Total Availability`, `Σ Total Demand`, and `Σ Total Number of Days`. The 'Demand/Availability Data' table is also shown with fields: `Σ availability`, `Σ demand`, `Order_Date_DD_MM_YYYY`, `Σ product_id`, `product_name`, and `Σ unit_price`. The report canvas displays a 'Data Visualization for Business Optimization' dashboard with three visual placeholders: 'Average Demand per Day', 'Average Availability Per Day', and 'Total Supply shortage'.

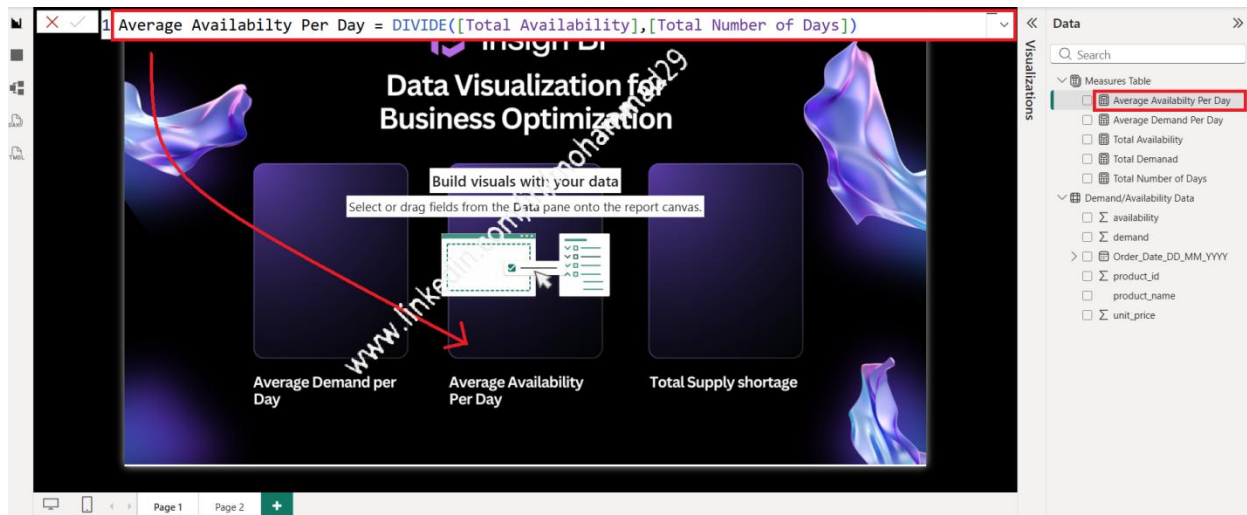
Day 7 - Power BI for Data Analysis

Step 3: We will build the KPI's for Page 1 using Measures.

i. We will make the first KPI i.e 'Average Demand per Day' using the Measure 'Total Demand' and 'Total Number of Days'.



ii. Now we will make the second KPI i.e 'Average Availability per Day' using the Measure 'Total Availability' and 'Total Number of Days'.

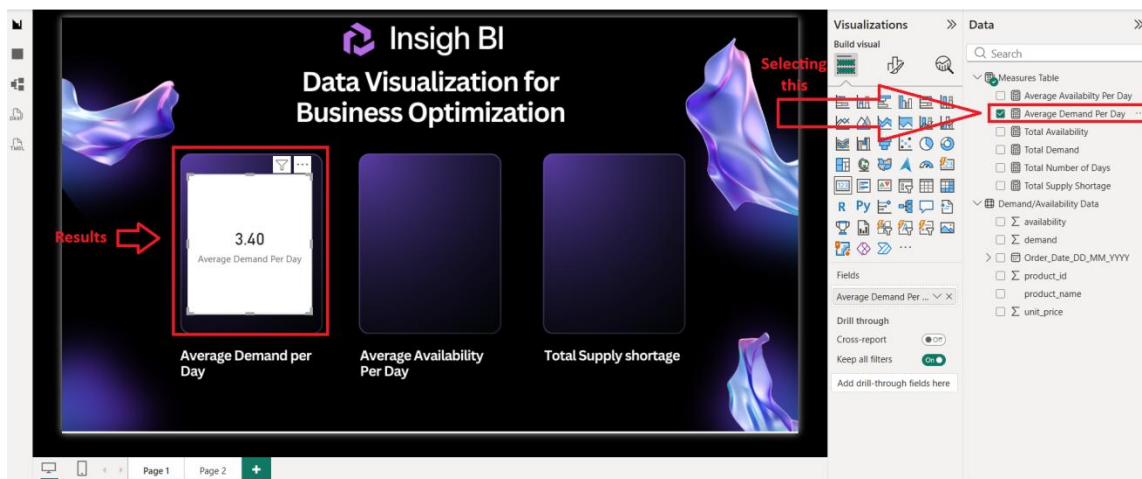
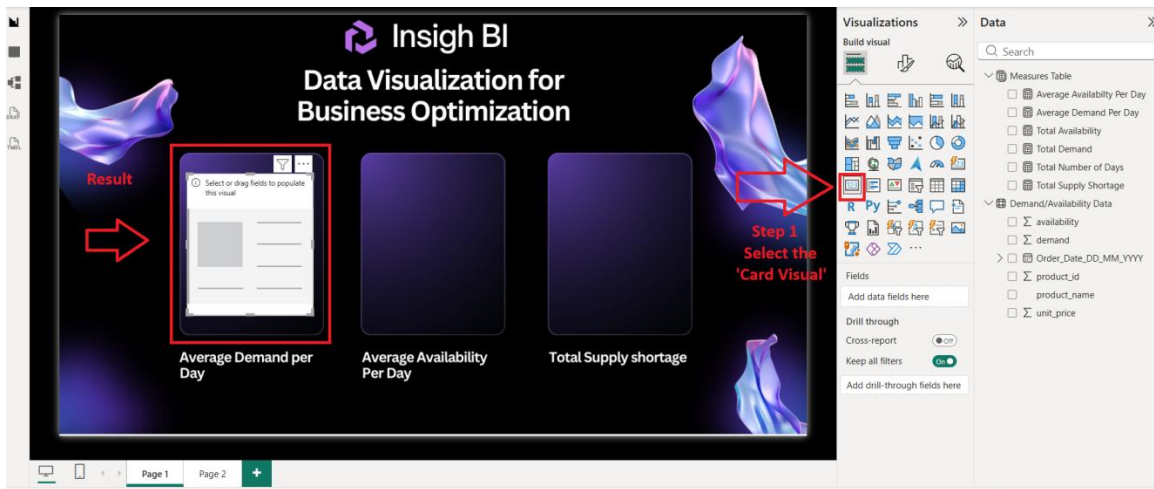


iii. Similarly we will find the third KPI i.e 'Supply Shortage'.

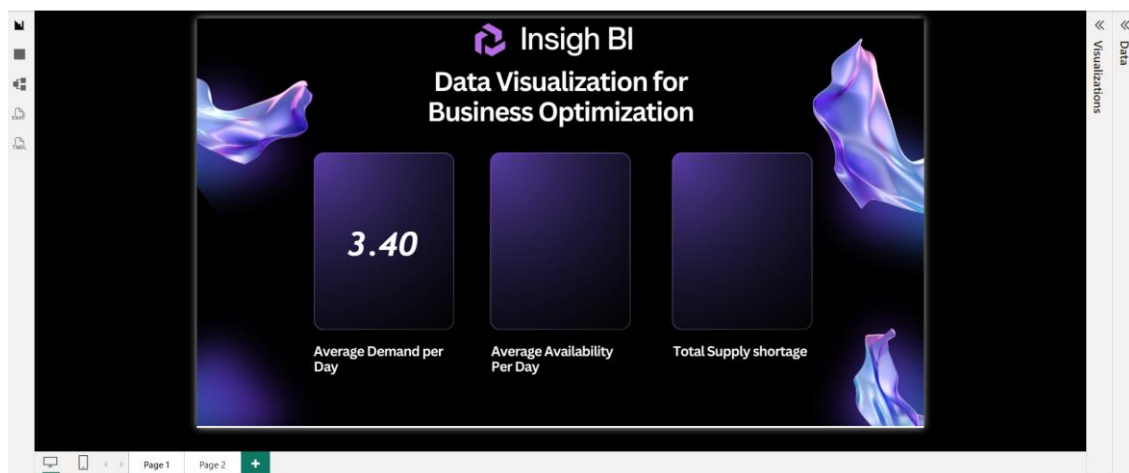
Day 7 - Power BI for Data Analysis

Step 4: Now we will make the card visuals for Page 1 using our Measures.

i. For the 'Average Demand Per Day'.

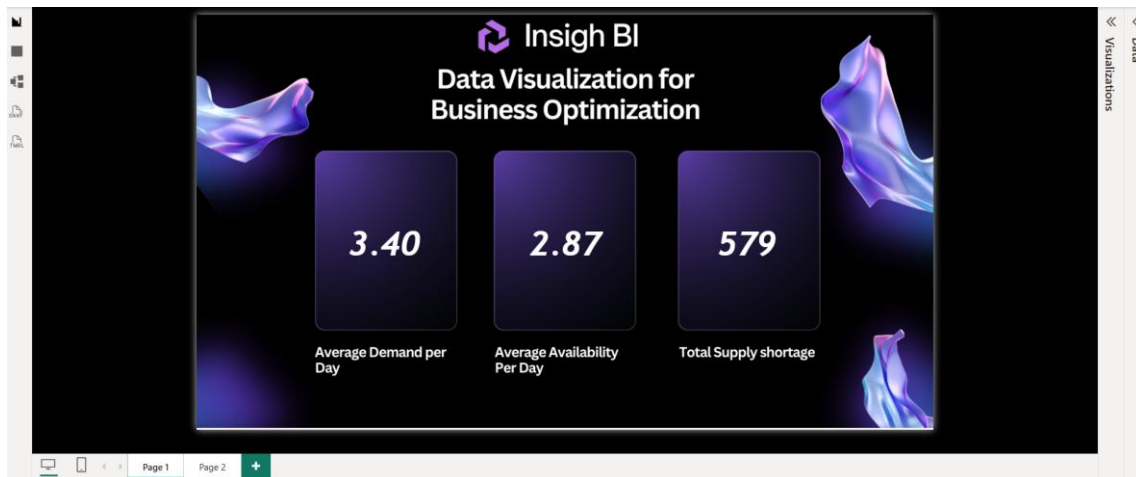


After Basic Editing, it will look like this



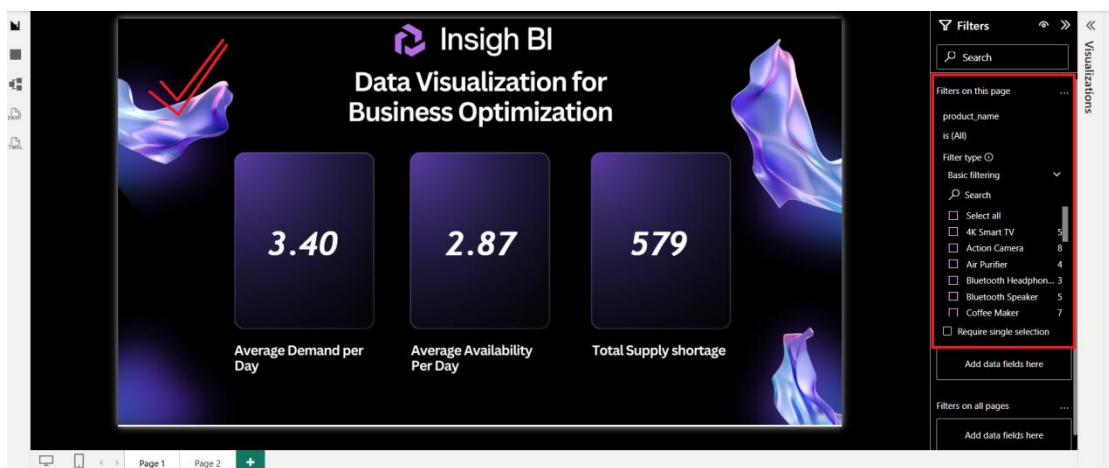
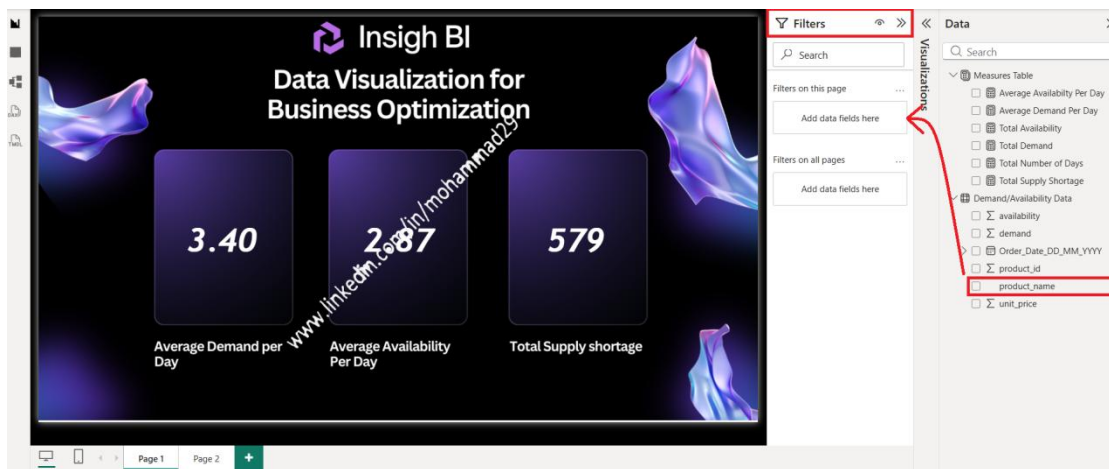
Day 7 - Power BI for Data Analysis

ii. Similarly we will do for others.



iii. To give flexibility to the user, we can make use of FILTER PANE.

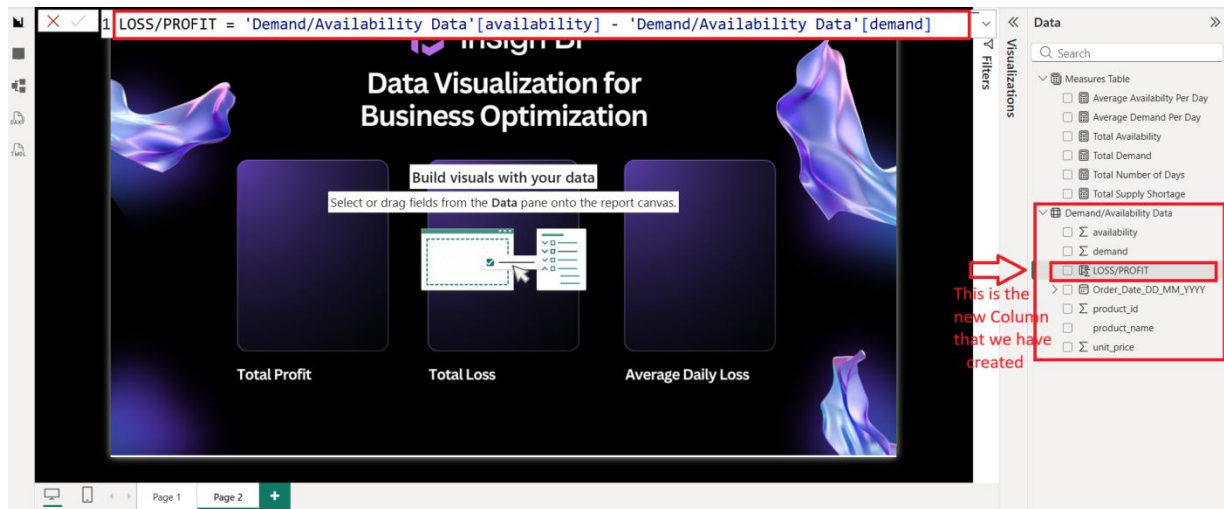
For example: Filtering the KPI based on the 'Product name'.



Day 7 - Power BI for Data Analysis

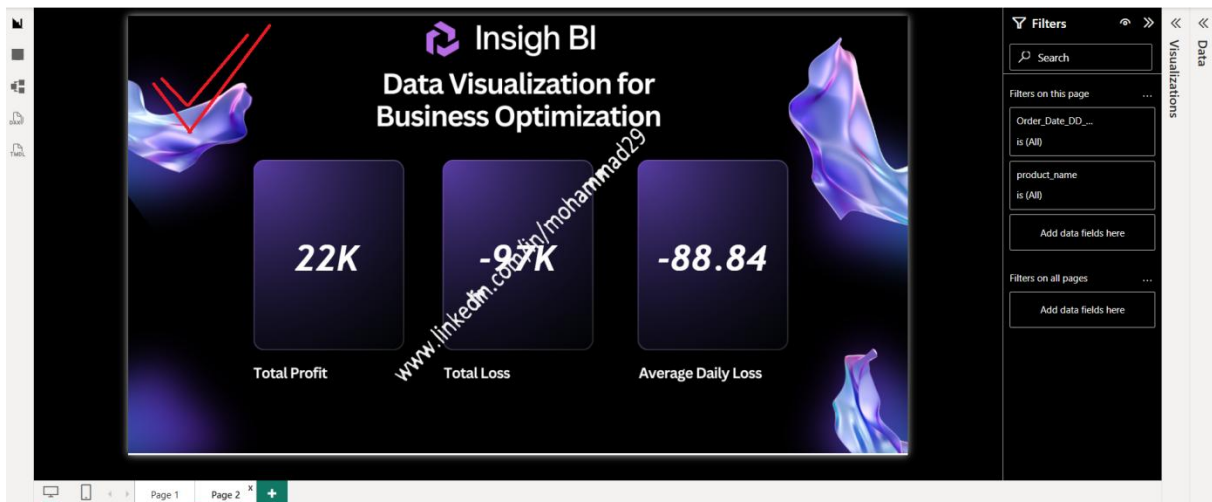
Step 5: Similarly KPI's and Card Visuals for Page 2

i. We will make a New column first i.e LOSS/PROFIT.



ii. After that we will create the Measures like 'Total Profit', 'Total Loss' and 'Average Daily Loss'.

iii. Finally we will create the Card visuals.



- Production Environment

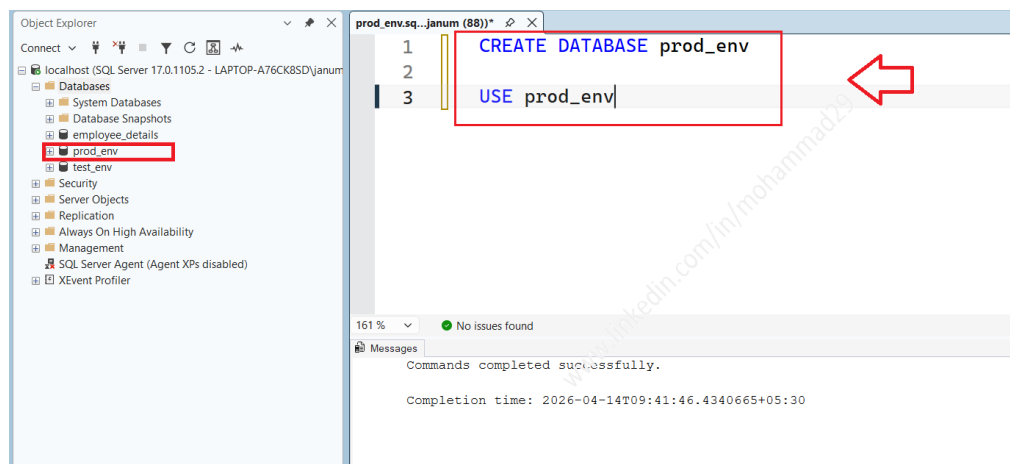
i. Test Dataset

We have few tables:

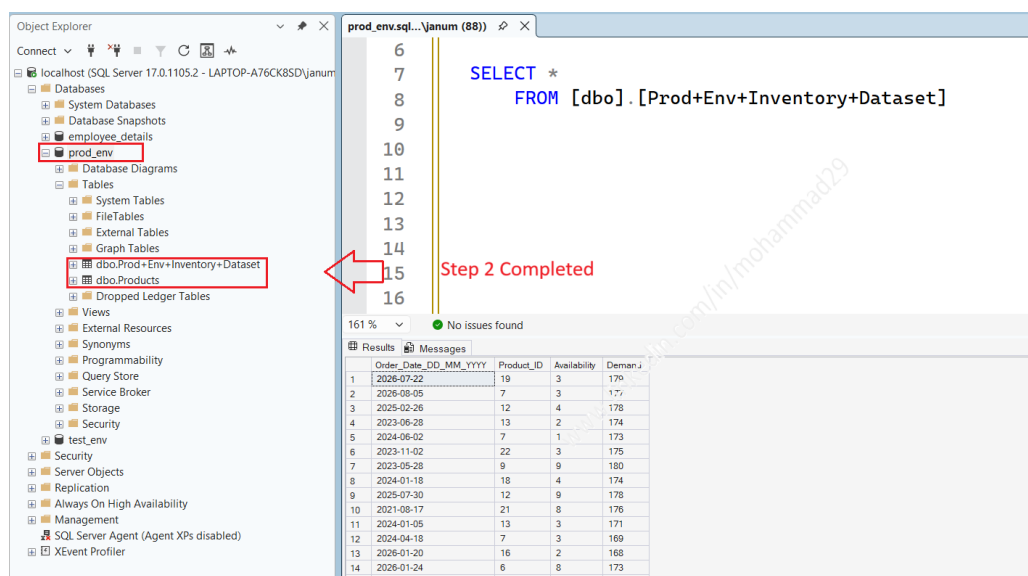
1. Demand & Availability Table
2. Products Table
3. KPI Sheet (that we have to represent on the PowerBI report)

ii. Importing the Production-Data in Production-Environment in SQL Server

Step 1: Create a new database by the name Prod_env database in SQL Server.



Step 2: Once the Database is created, import the test-data in the test_env.



Day 7 - Power BI for Data Analysis

Step 3: Data Analysis & Cleaning.

Step 4: Applying Left Join in SQL Server.

Products Table

	Product_ID	Product_Name	Unit_Price
1	1	Wireless Mouse	15.9899997711182
2	2	Bluetooth Headphones	39.9900016784668
3	3	4K Smart TV	499.989990234375
4	4	Portable Charger	25.4899997711182
5	5	Gaming Keyboard	79.9899978637695
6	6	Laptop Stand	29.9899997711182
7	7	Phone Case	12.9899997711182
8	8	Digital Camera	299.989990234375
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10	10	Tablet	249.990005493164
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12	12	Electric Toothbrush	49.9900016784668
13	13	Air Purifier	119.98999786377
14	14	Coffee Maker	59.9900016784668

Prod+Environment+Inventory+Dataset

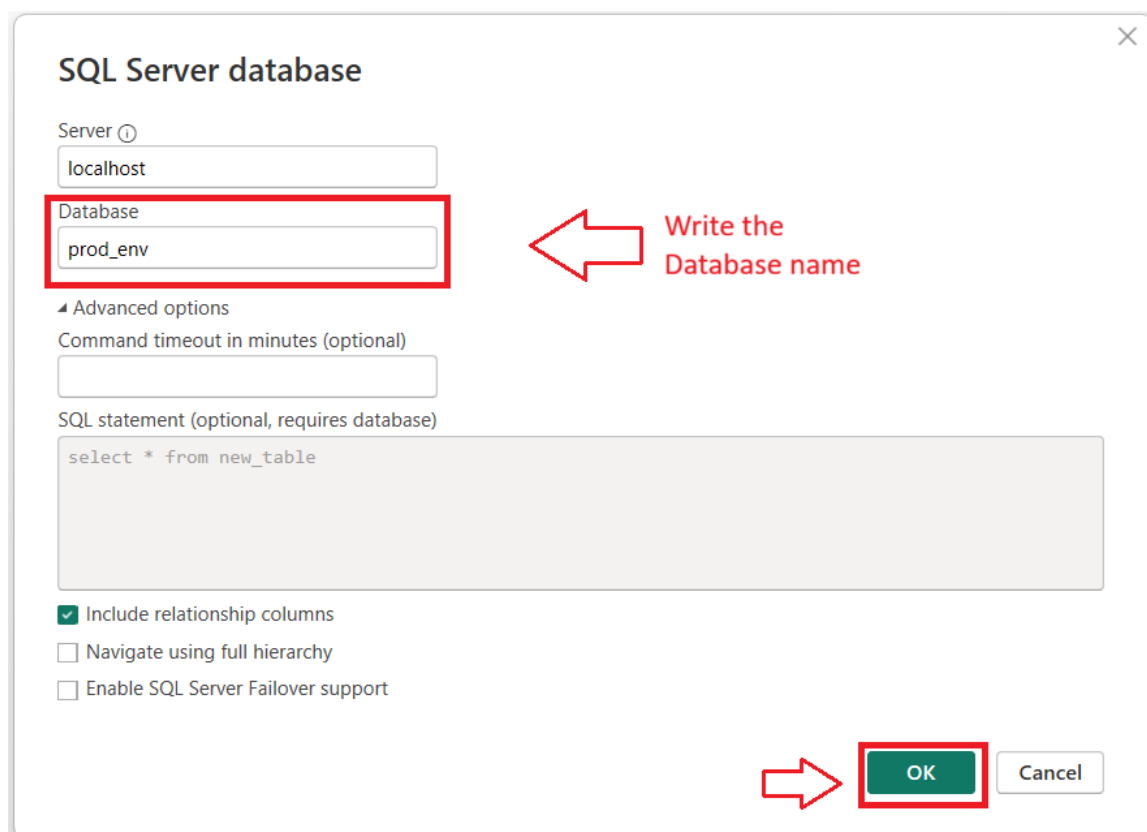
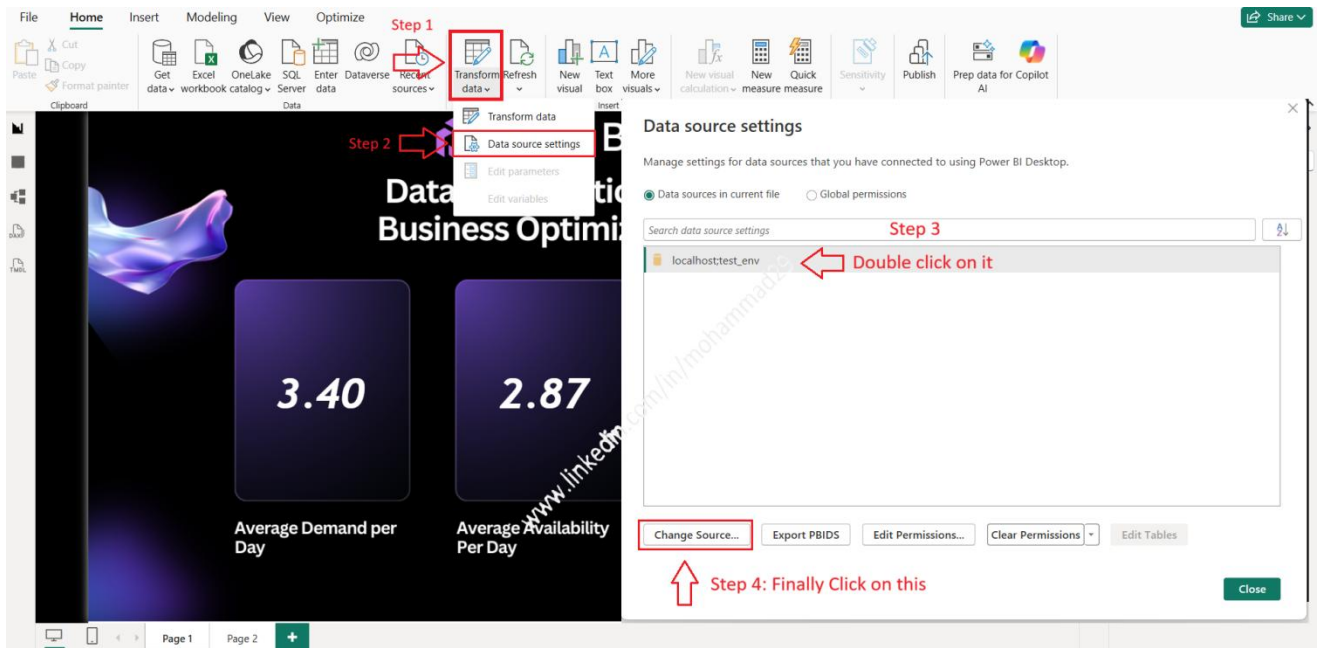
	Order_Date_DD_MM_YYYY	Product_ID	Availability	Demand
1	2024-11-11	18	31	43
2	2024-11-13	17	38	50
3	2025-10-28	14	34	46
4	2025-10-14	11	38	50
5	2025-03-20	13	37	49
6	2024-05-16	20	25	37
7	2025-05-26	17	26	38
8	2025-05-01	18	37	49
9	2024-02-09	2	29	41
10	2025-10-30	19	35	47
11	2025-06-08	11	0	12
12	2025-07-10	2	37	48
13	2026-12-06	15	38	49
14	2026-01-11	8	26	37

Left Join

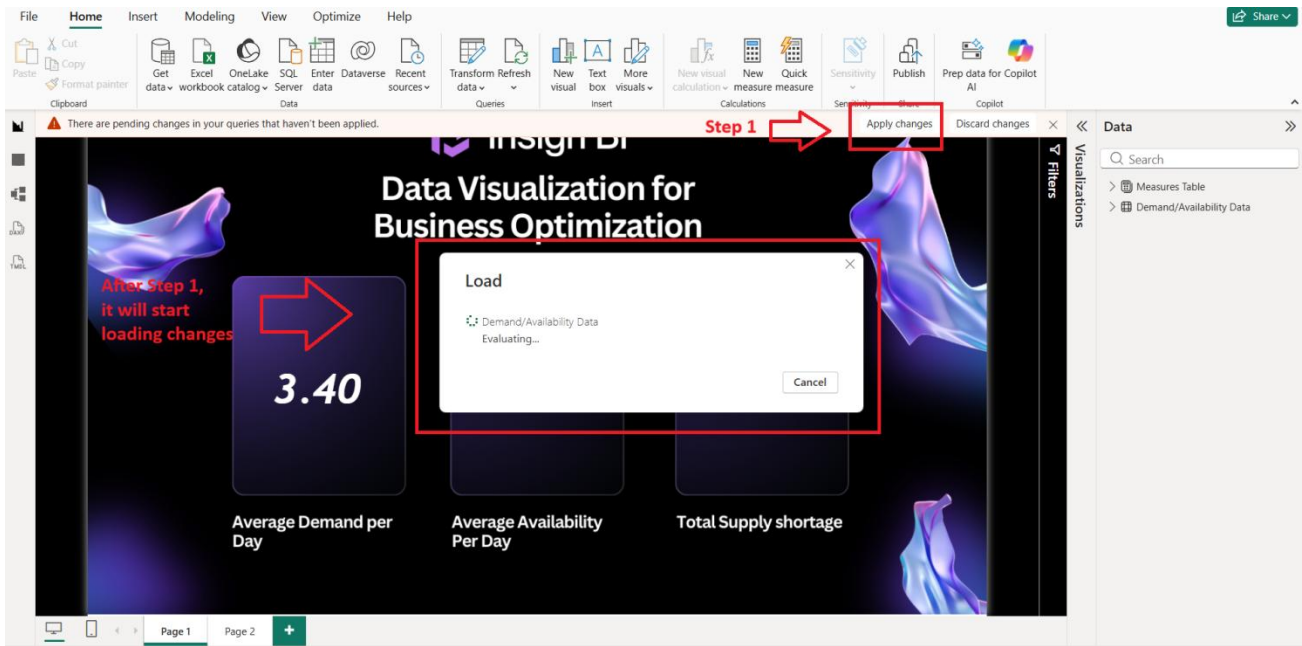
new_table

	Order_Date_DD_MM_YYYY	Product_ID	availability	demand	product_name	unit_price
1	2024-11-11	18	31	43	VR Headset	399.989990234375
2	2024-11-13	17	38	50	Action Camera	169.990005493164
3	2025-10-28	14	34	46	Coffee Maker	59.9900016784668
4	2025-10-14	11	38	50	Noise-Canceling Headphones	89.9899978637695
5	2025-03-20	13	37	49	Air Purifier	119.98999786377
6	2024-05-16	20	25	37	Smart Thermostat	129.990005493164
7	2025-05-26	17	26	38	Action Camera	169.990005493164
8	2025-05-01	18	37	49	VR Headset	399.989990234375
9	2024-02-09	2	29	41	Bluetooth Headphones	39.9900016784668
10	2025-10-30	19	35	47	Bluetooth Speaker	39.9900016784668
11	2025-06-08	11	0	12	Noise-Canceling Headphones	89.9899978637695
12	2025-07-10	2	37	48	Bluetooth Headphones	39.9900016784668
13	2026-12-06	15	38	49	LED Desk Lamp	18.9899997711182
14	2026-01-11	8	26	37	Digital Camera	299.989990234375

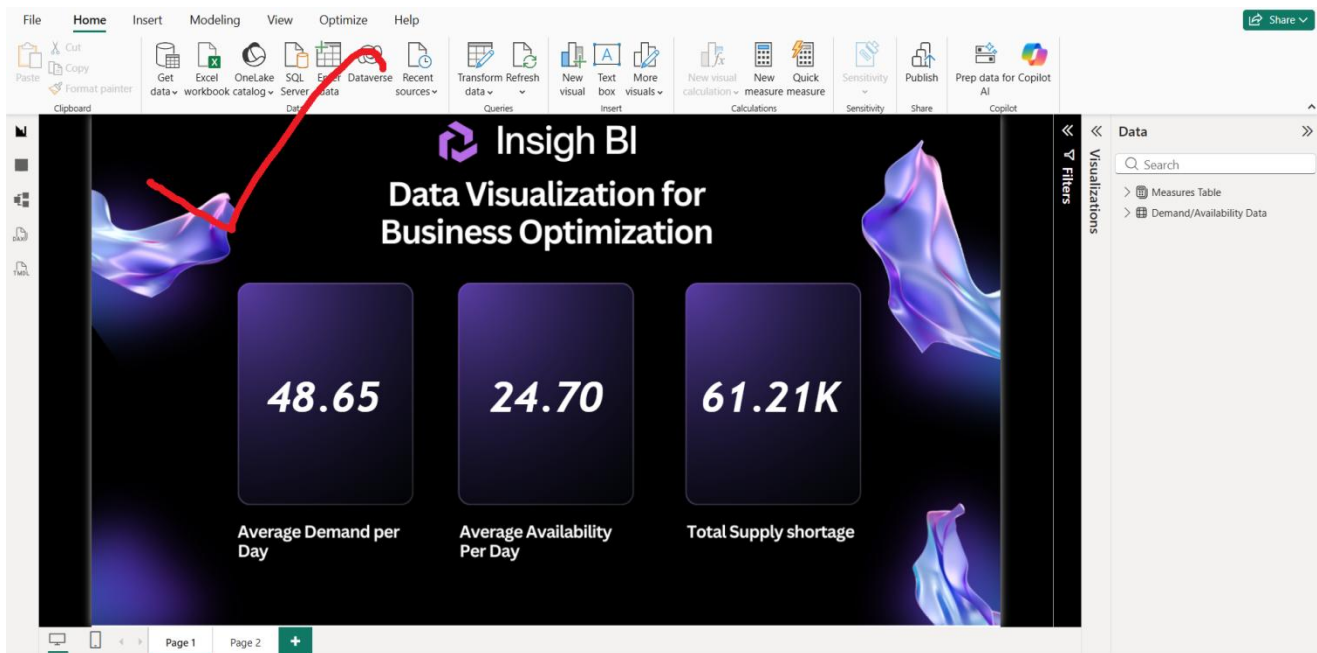
iii. Switching the Test-Env to Prod-Env by importing the New-Table of Test-Env in SQL Server to Power BI Desktop



Day 7 - Power BI for Data Analysis



Finally we can see the updated report.



3. Connecting Power BI with MySQL Database

3.1. Download MySQL Database

Install both '**MySQL Workbench**' as well as '**MySQL Server**'.

- MySQL Workbench is a GUI tool to run queries and manage databases.
- MySQL Server must be installed and running for Workbench to connect to it.

Steps to install on Windows:

LINK: <https://dev.mysql.com/downloads/installer/>

1. Download the full (353 MB) Community version, not the shortcut or web-only version, to avoid missing essential components like MySQL server.
2. During installation, select 'Full' setup type.
3. Set the 'root username' and 'password' when prompted.
4. Complete installation and open MySQL Workbench.

NOTE: Issues during installation might require uninstalling previous versions and reinstalling with the full setup.

3.2. Plan of Action

Q. How can we change the data source & update the report without affecting DAX calculations or existing work?

Before making any changes, it is recommended to create a backup of the SQL Server report by publishing it to a Power BI account.

Steps:

- i. Log in to your Power BI account.
 - ii. **Before transitioning:**
First, publish the report (created using SQL Server) to a separate workspace in Power BI. This ensures that we have a backup copy of the report.
 - iii. Next, create another separate workspace where we will publish the report that uses the **MySQL database** as the data source.
 - iv. This helps ensure that if anything goes wrong during the transition or if any errors occur, we still have the original SQL Server-based report available. We can use it to make corrections if required.
- **Note:** If publishing is not possible, you can simply create a **copy** of the file on your local machine.

- Connecting Power BI with MySQL Workbench

- Use **MySQL Connector** to establish the connection.

- Changing Data source from 'SQL Server' → 'MySQL'

In Power BI

Steps:

- i. Go to **Transform Data**.
 - ii. Select the new query created using MySQL.
 - iii. Under the **Home** tab, open **Advanced Editor** and copy the query code.
 - iv. Now select the existing query that was imported from SQL Server.
 - v. Again, go to **Home** → **Advanced Editor**, and replace the existing code with the copied MySQL query code.
 - vi. Click **Close & Apply**.
- Now, publish this report as well, with MySQL as the data source.

- Important Observation

- The data remains the same; only the **data source changes**.
- Therefore, both reports should produce identical results.

- Note

Changing 'Test' → 'Production Environment' in Same Database

You can transition the report source in two ways:

1. Transform Data → Data Source Settings
2. Power Query Editor

Changing (Switching) from 'one Database' → to 'another Database'

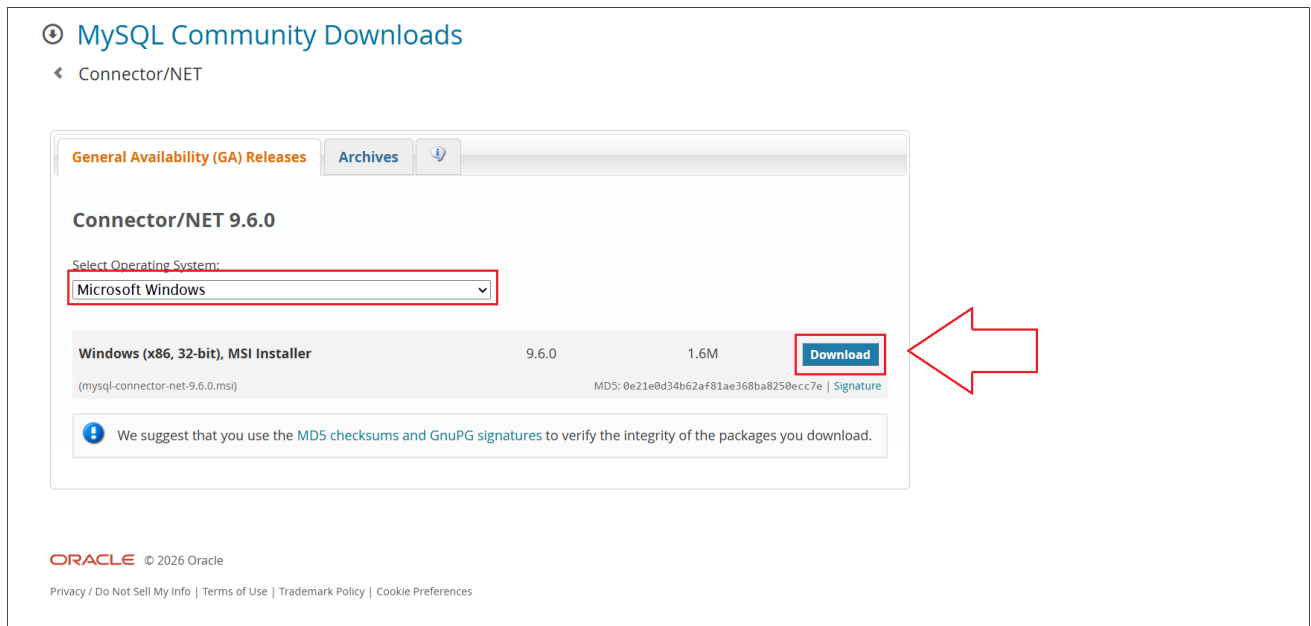
1. Will need to use the **Advanced Editor** to modify the query accordingly.

3.3. Practical – MYSQL

i. Downloading MYSQL Connector

We use **MySQL Connector** to establish the connection of Power BI with MYSQL Workbench.

LINK: <https://dev.mysql.com/downloads/connector/net/>



- Production Environment

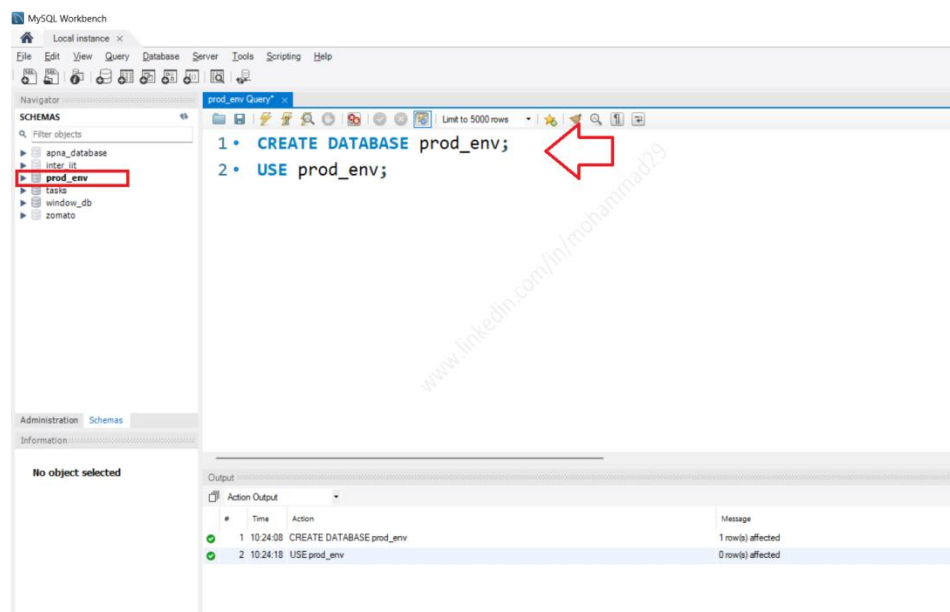
i. Production Dataset

We have few tables:

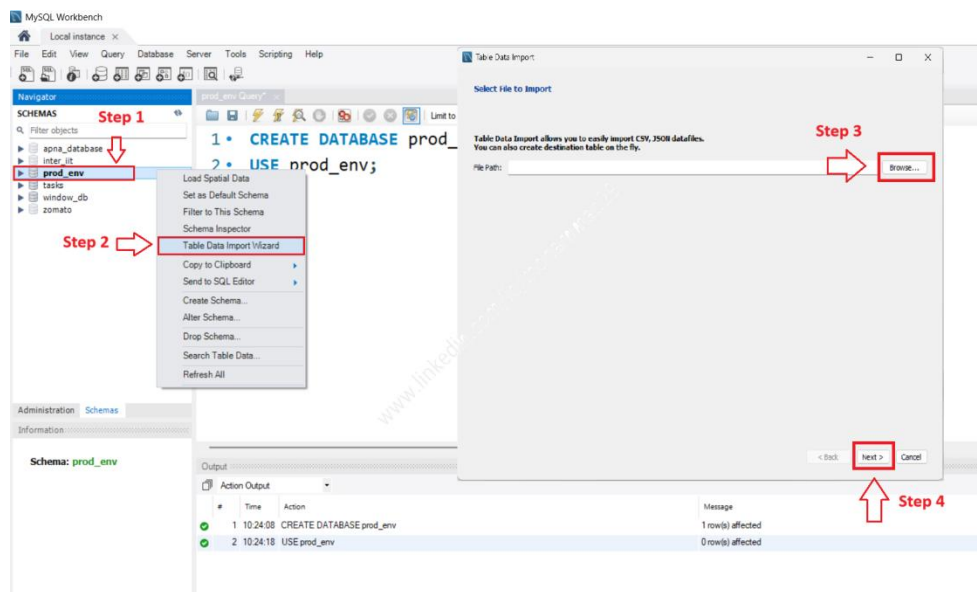
1. Demand & Availability Table
2. Products Table

ii. Importing the Prod-Data in Prod-Environment in MySQL Database

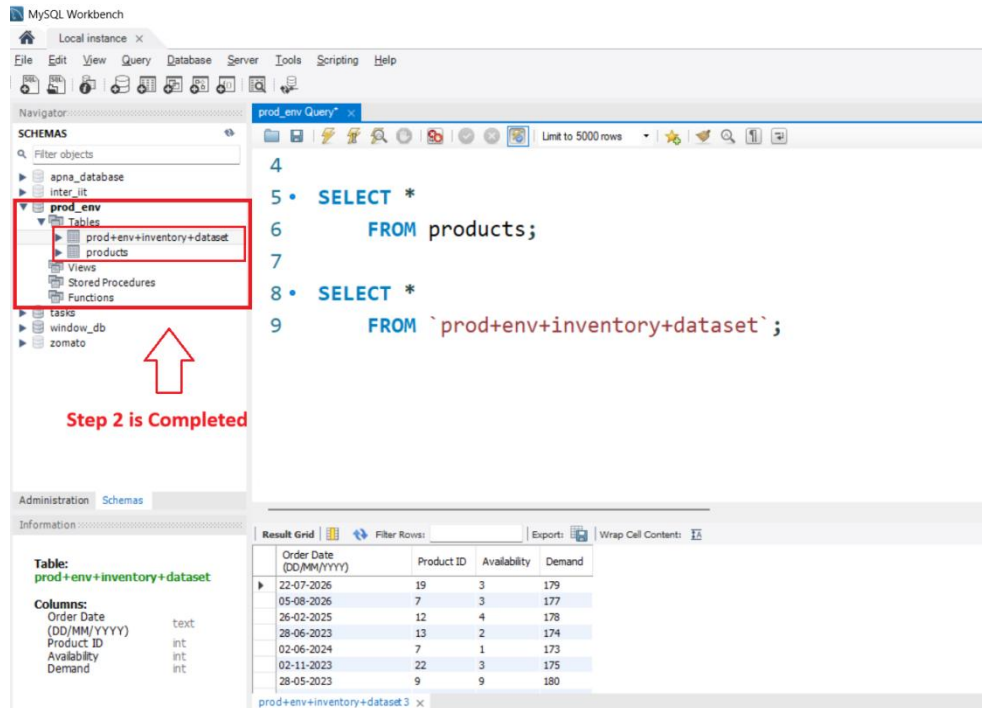
Step 1: Create a new database by the name prod_env database in MySQL Database.



Step 2: Once the Database is created, import the prod-data in the prod_env.

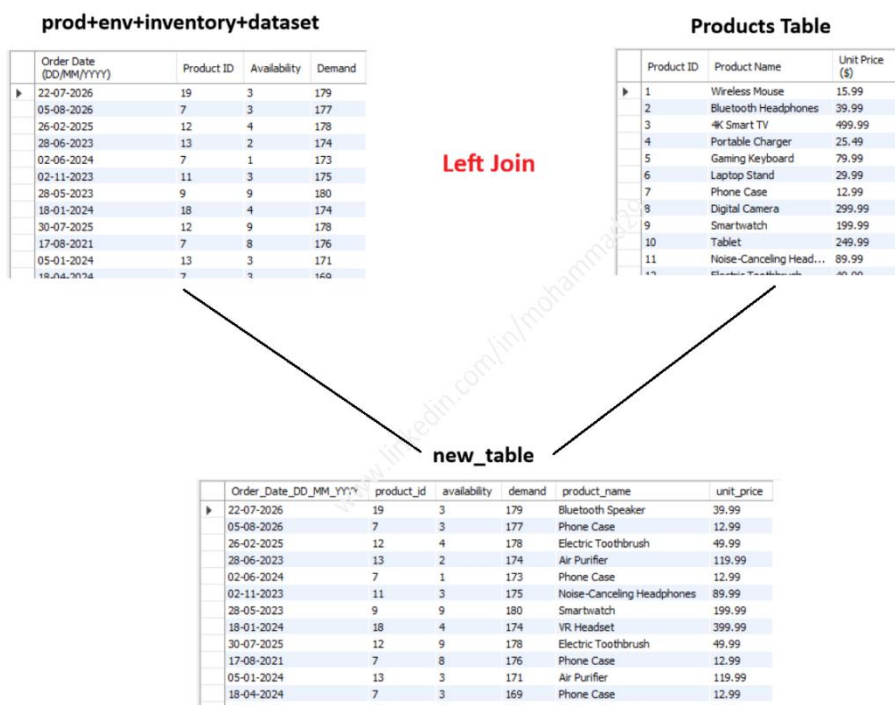


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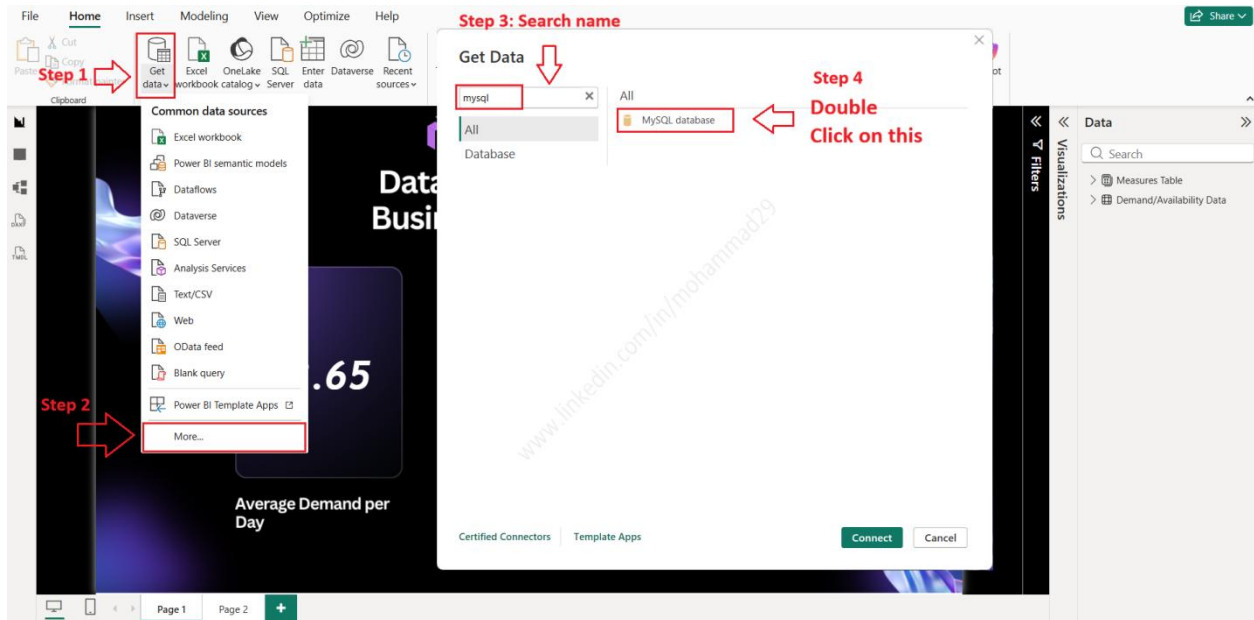
Step 3: Applying a LEFT JOIN in the MySQL database to prepare the data for reporting.

Note: Here, we will use the same SQL Server code, but adapt it to MySQL syntax while keeping the column names consistent with the SQL Server version. This ensures there are no issues during reporting, as we will only be changing the database, and the DAX will remain the same.

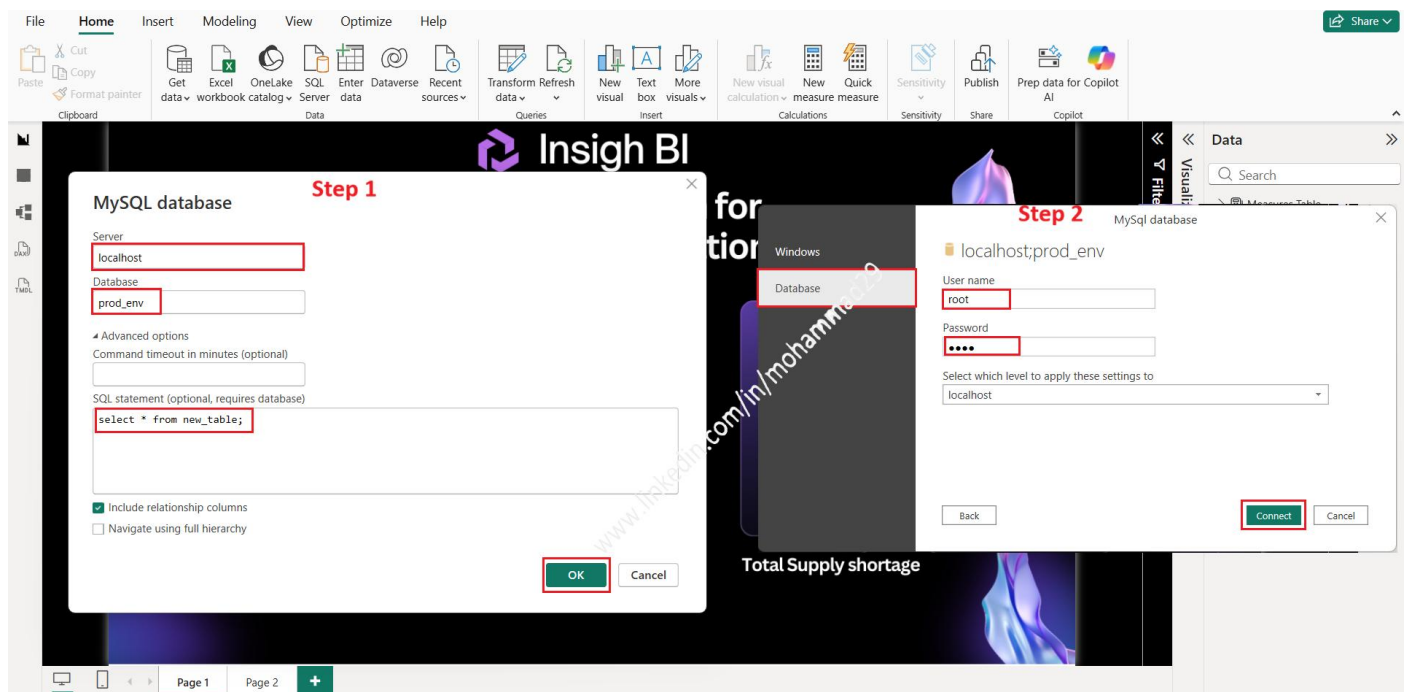


iii. Importing the New-Table to Power BI Desktop from in Production Environment

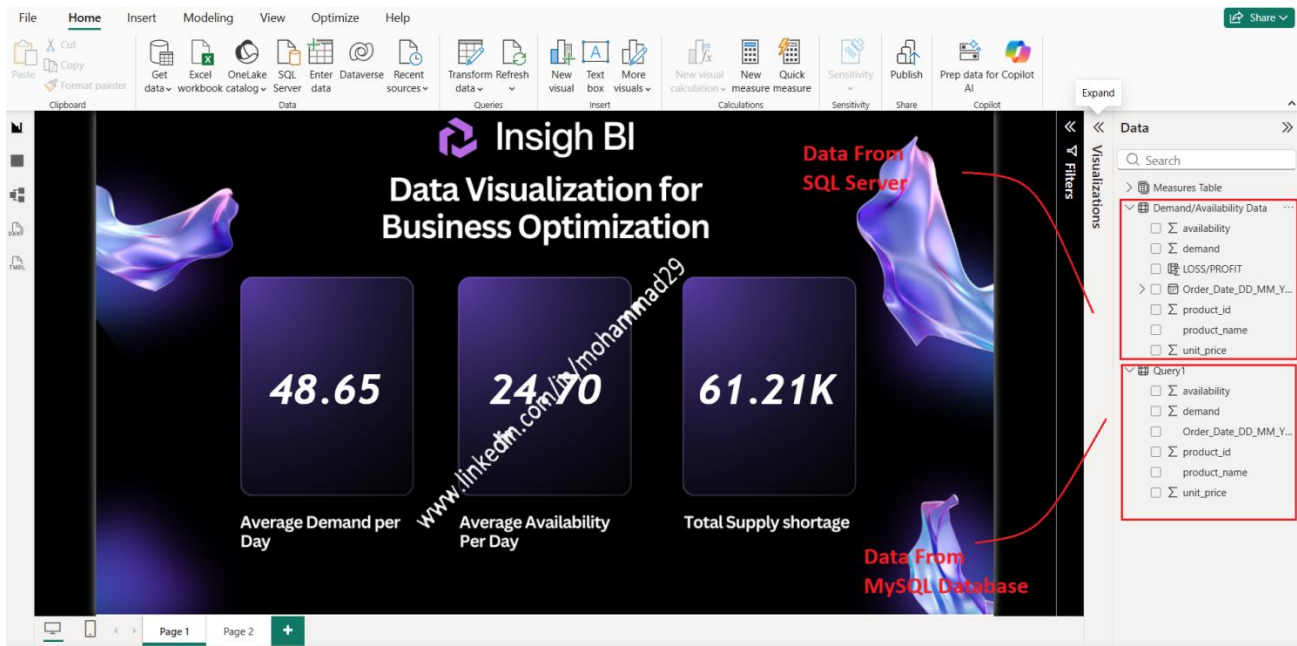
Step 1: Open Power Bi Desktop and select SQL Server as the data source.



Step 2: Connect MySQL Database to Power BI and Load the production Data.



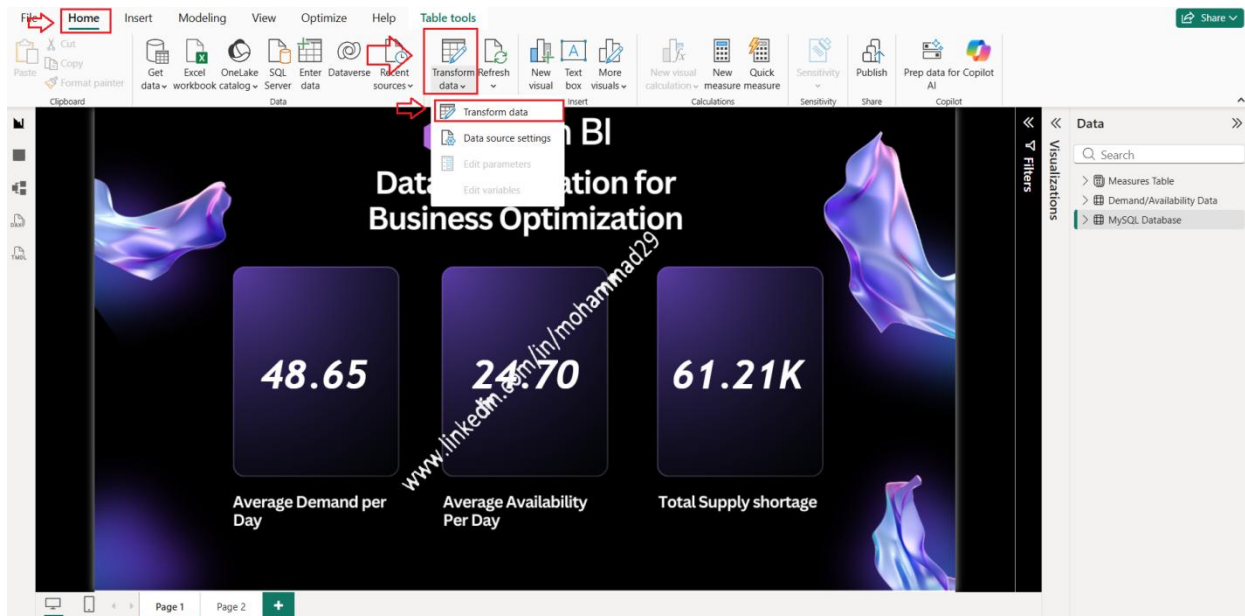
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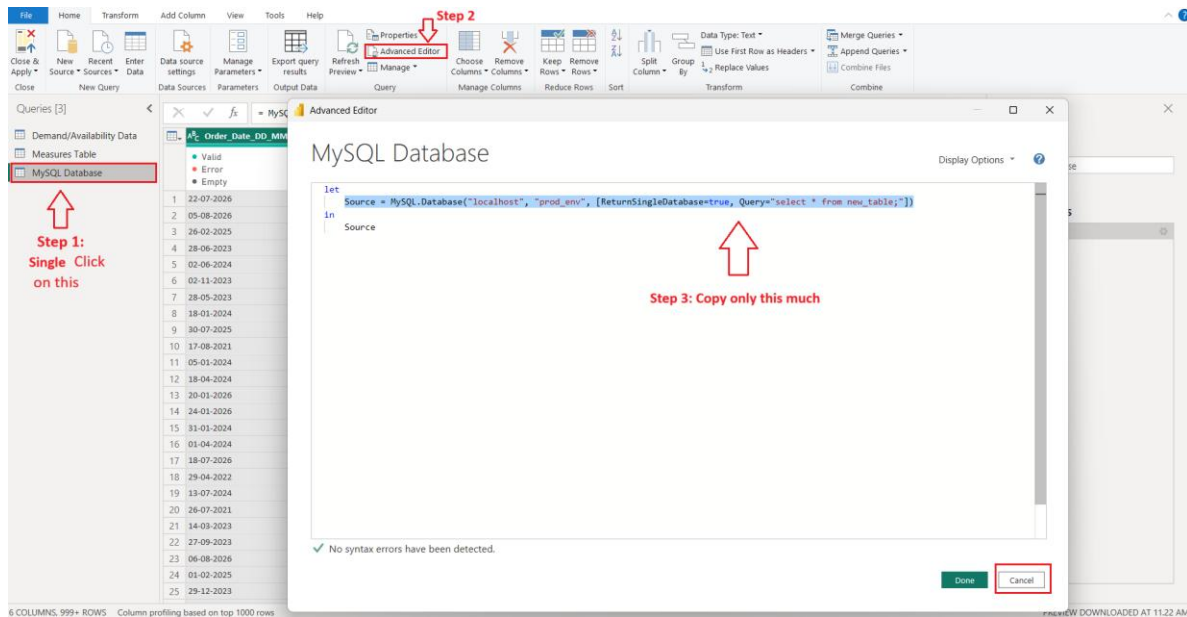
iv. Transitioning the production report from SQL Server as the data source to MySQL database.

Changing (Switching) from 'one Database' → to 'another Database'

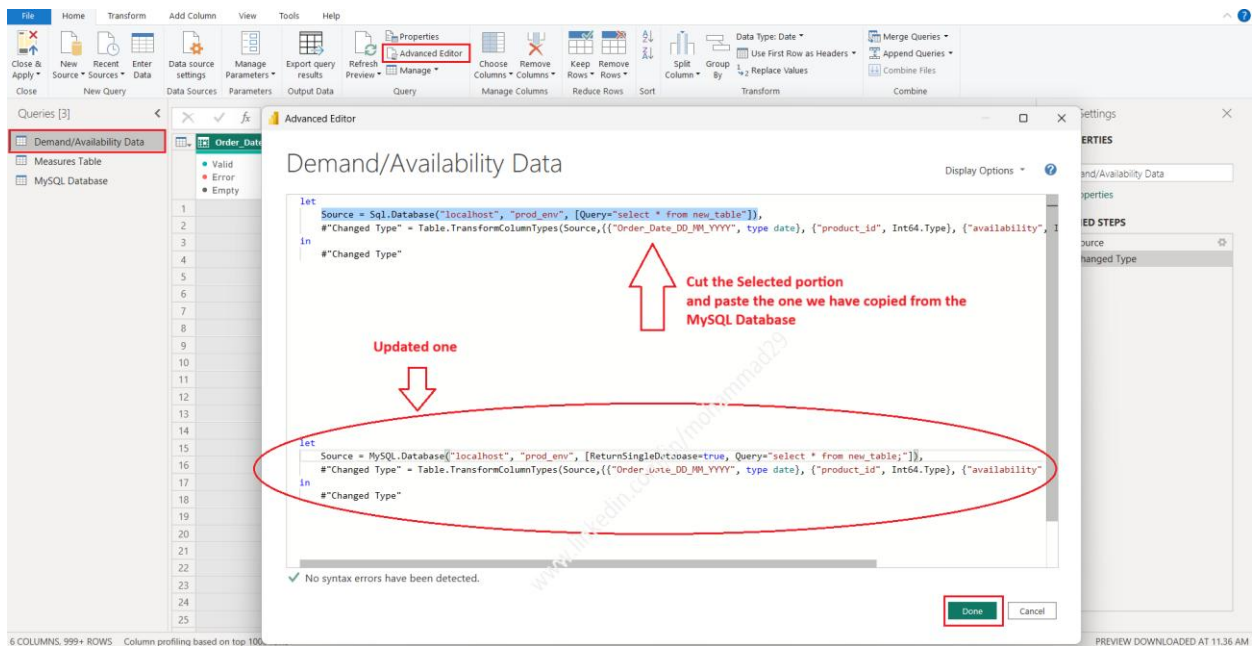
1. Will need to use the **Advanced Editor** in Power Query Editor to modify the query accordingly.



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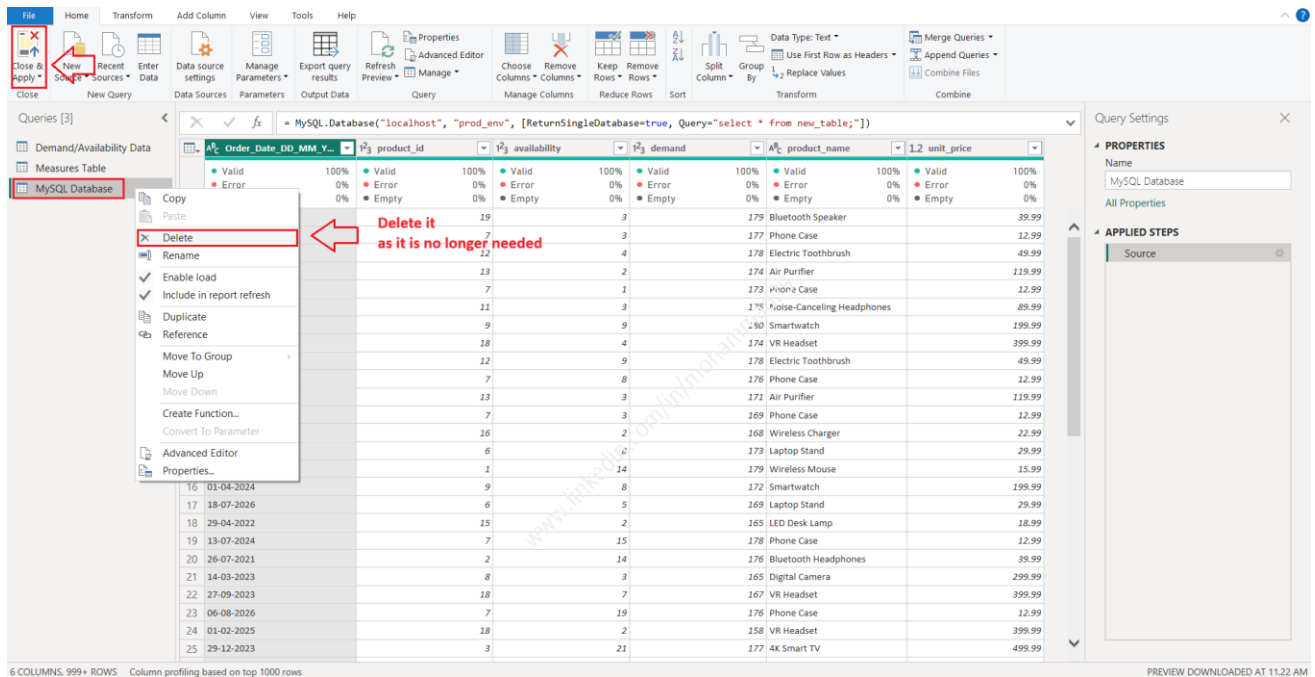


Now we will change the data source of the 'Demand/Availability Data' from SQL Server to MySQL.

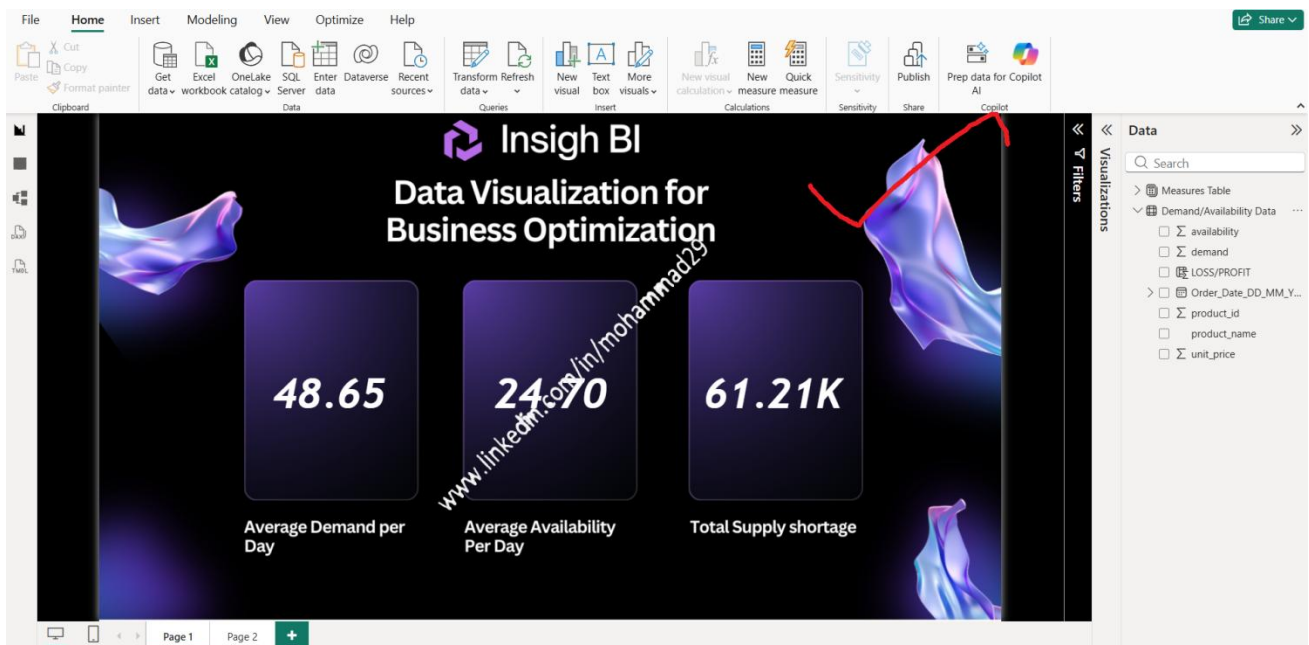


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Now I will Delete the 'MySQL Data' and Click on 'Close & Apply'.



Now the Data source is 'MySQL Database'.

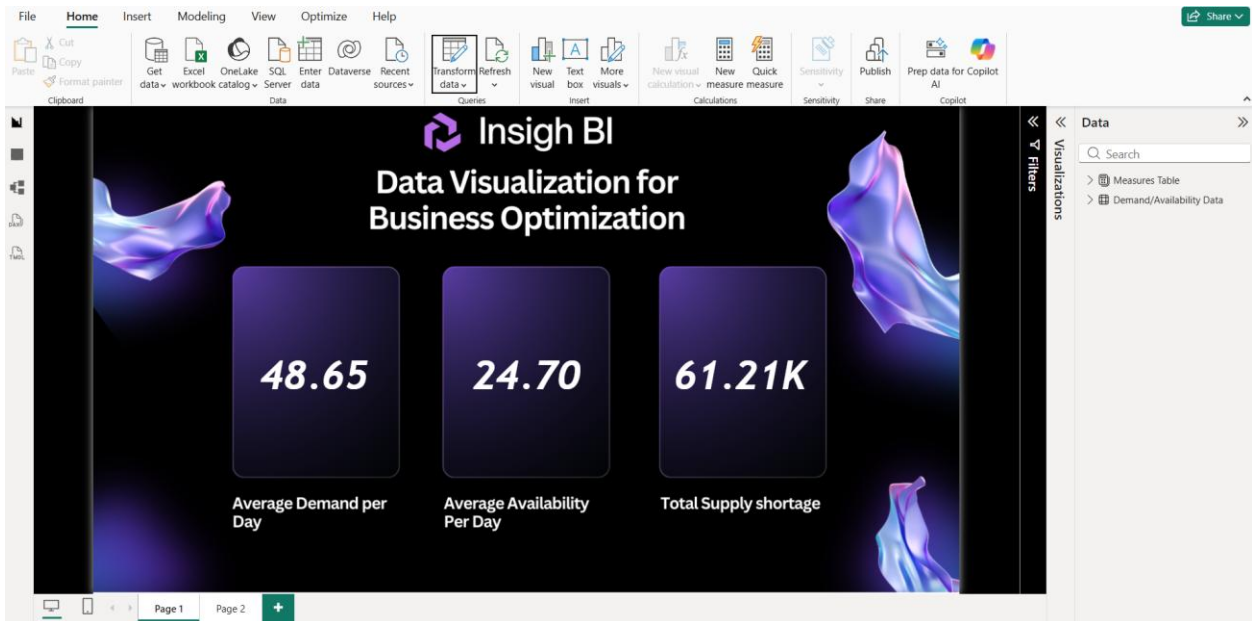


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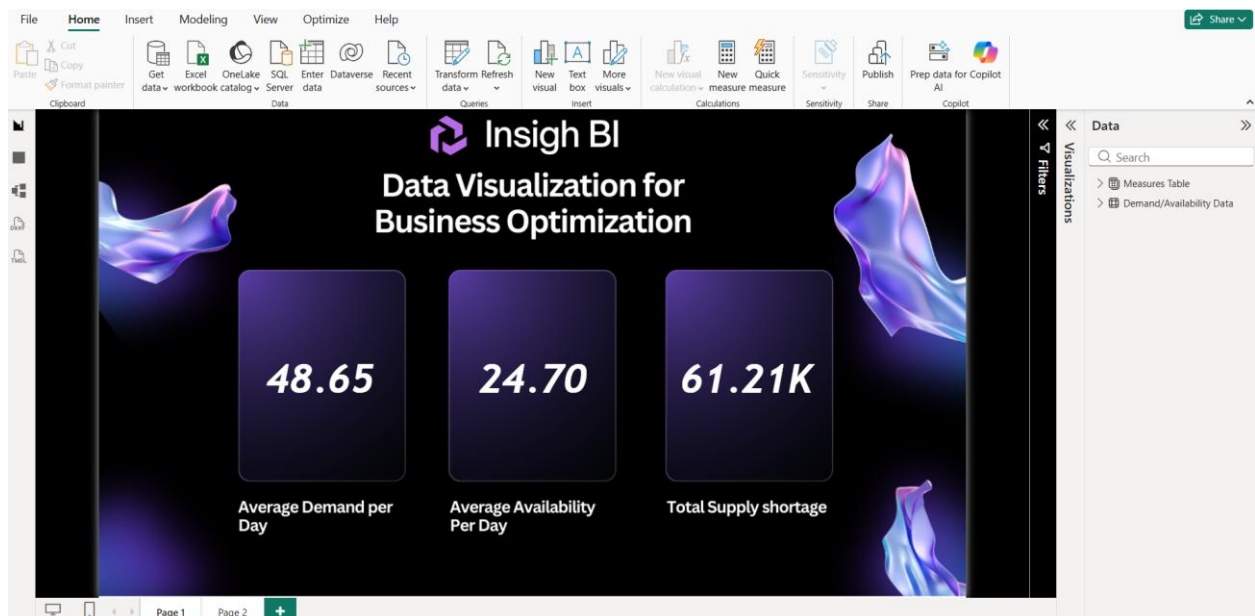
v. Data Validation

The numbers shown in this report (with MySQL as the data source) must match those published/stored in the report that used SQL Server as the data source.

The report below uses MySQL as its data source.



Now, we will verify the report data that previously used SQL Server as the data source.



Since both reports contain the same data, we were able to switch the report's data source from one database to another without affecting the DAX calculations.